

ShiroFOC hardware design

Schematic topology and component decisions

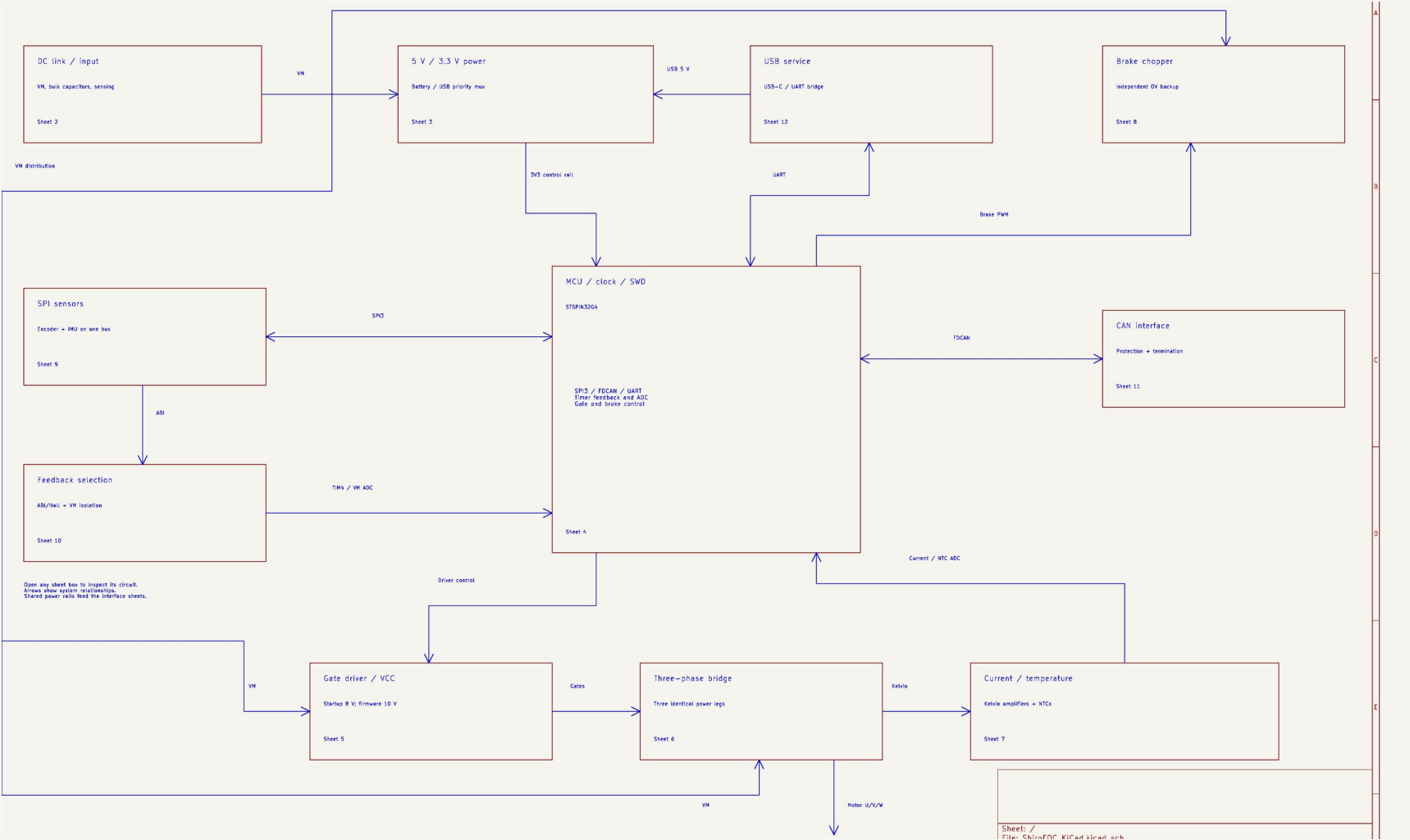
A single-axis FOC servo module

Topology, component choices and the measurements that establish the operating envelope.

A1 drawing / A0 electrical design

16 September 2026

36 review slides + 5 technical appendices



Prototype targets. Hardware qualification remains open.

Electrical requirements and operating envelope

18–42 V bus

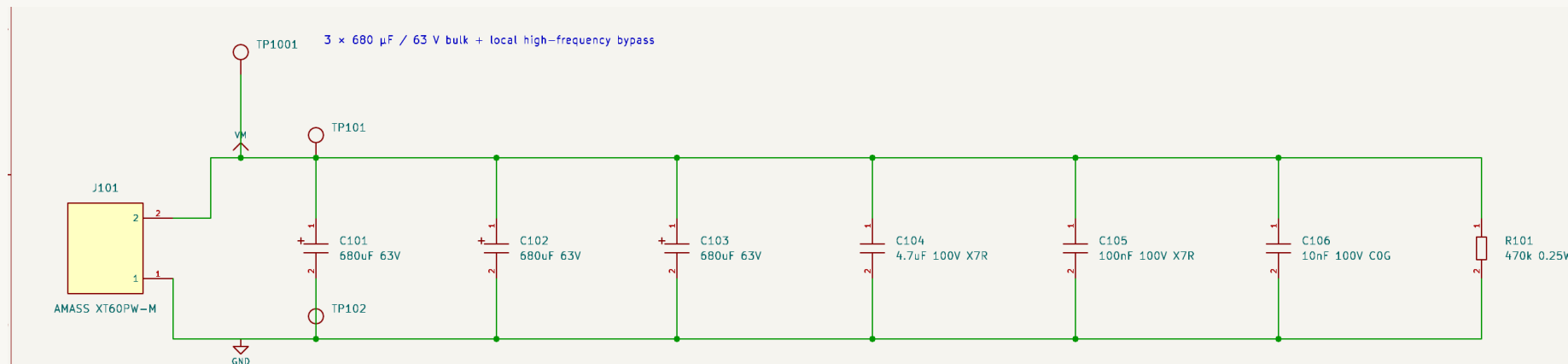
6S–10S packs. The preferred design point is 8S: 29.6 V nominal, 33.6 V fully charged.

25 / 40 Arms targets

Passive cooling / a defined thermal solution. Phase current differs from battery current.

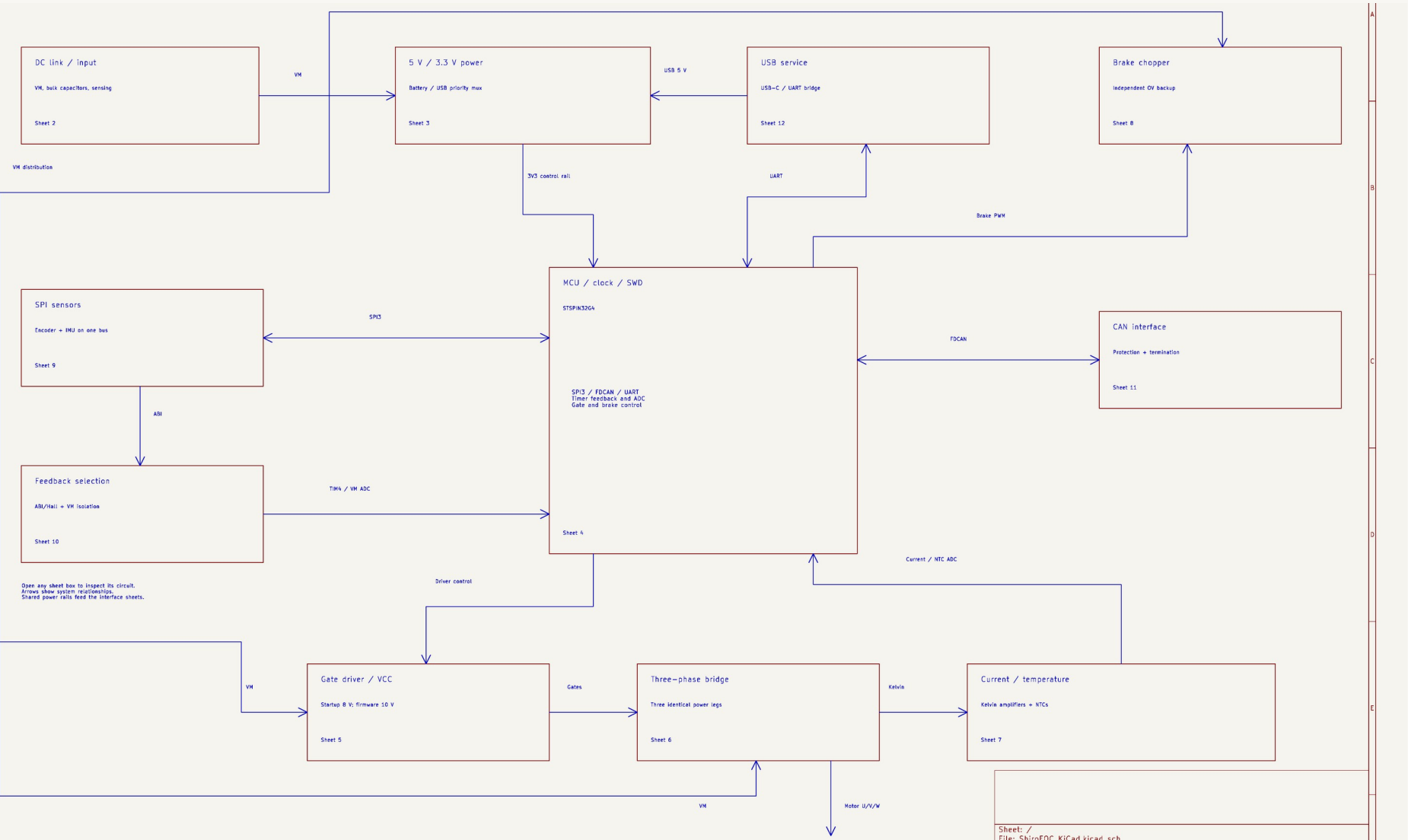
80 A peak is a future target

At most 2 s, subject to qualification. Initial +60 A protection prevents treating this as an enabled rating.



60 V MOSFET absolute maximum requires margin for regeneration and switching overshoot.

System architecture and FOC signal flow



Local current control

Shunts measure phase current. The MCU uses rotor position and current feedback to calculate PWM.

Outer motion loops

Velocity and position loops set the requested torque-producing current. CAN coordinates modules.

Integrated control, external power

U301 combines the MCU, driver and analog resources. External MOSFETs set power-stage capability.

Position feedback ♦ phase current ♦ synchronized PWM form the local control loop.

Power domains and operating modes

Supply state	Logic	Power stage
Battery only	5V_BAT selected	Available after setup
USB only	5V_USB selected	VM / VCC absent
Both valid	USB preferred	Available after setup
VM returns	MCU may stay alive	Reinitialize before arming

Control power survives USB service

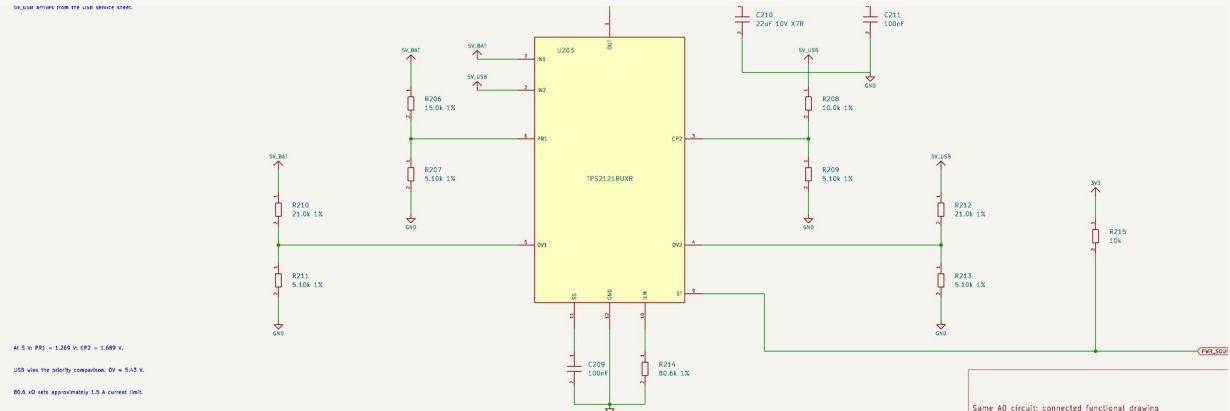
5V_BAT and 5V_USB feed TPS2121.
5V_SYS feeds the external 3.3 V LDO.

Gate power follows VM

The internal STSPIN buck generates VCC independently of the 3.3 V control rail.

VM return requires reinitialization

USB can keep the MCU alive while the gate driver loses power. Restore VCC settings and protections before torque.

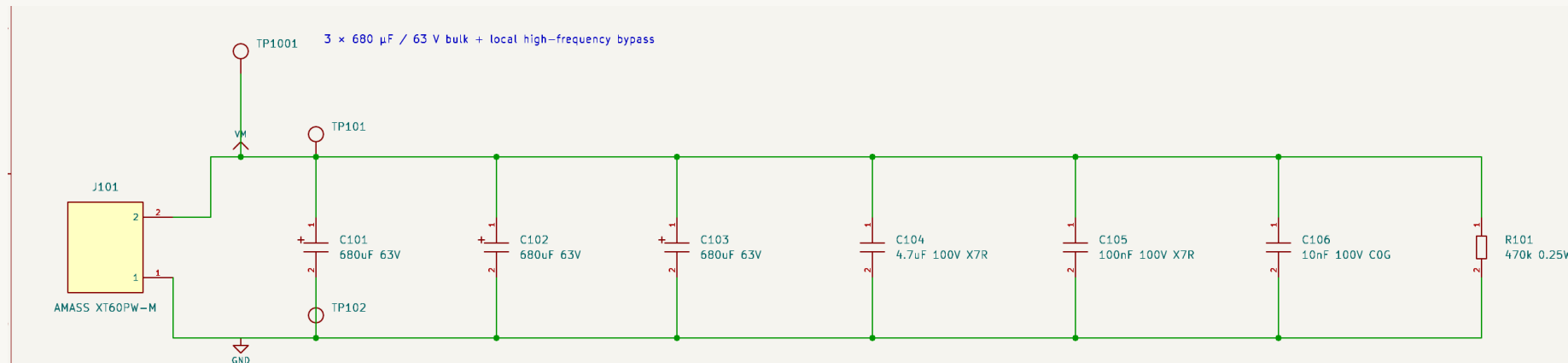


USB alone powers logic and sensors. It does not energize VM, VCC or either power switch stage.

Battery entry and DC-link capacitance

2040 μF bulk capacitance

C101–C103 are 680 μF / 63 V each. The bank buffers short bus-energy changes.



Several capacitor scales

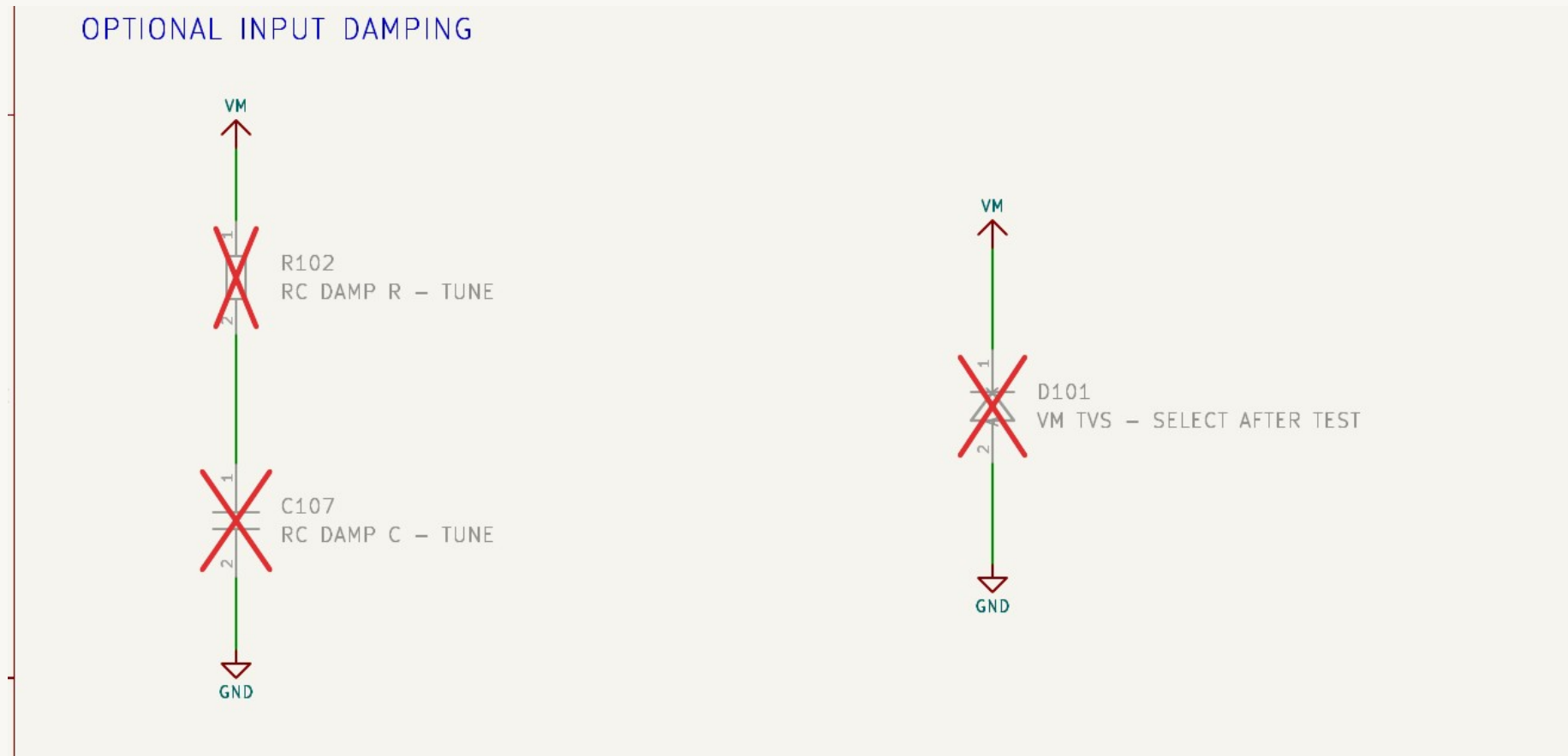
4.7 μF , 100 nF and 10 nF bypass paths support progressively faster current changes.

Ripple rating can be limiting

5.07 Arms is the summed 100 kHz nameplate rating. Actual sharing, frequency and temperature change the usable margin.

$$E = \frac{1}{2}CV^2 = 1.80 \text{ J at } 42 \text{ V}$$

Inrush, discharge and input damping



Bidirectional battery path

Regenerated current can return to a suitable battery. Rev A omits reverse-polarity protection and active precharge.

470 kΩ is a slow bleeder

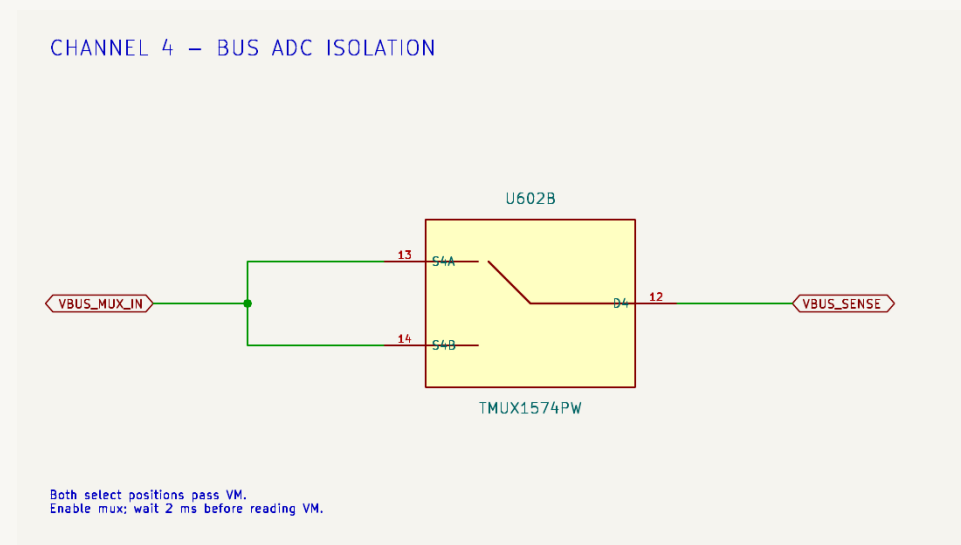
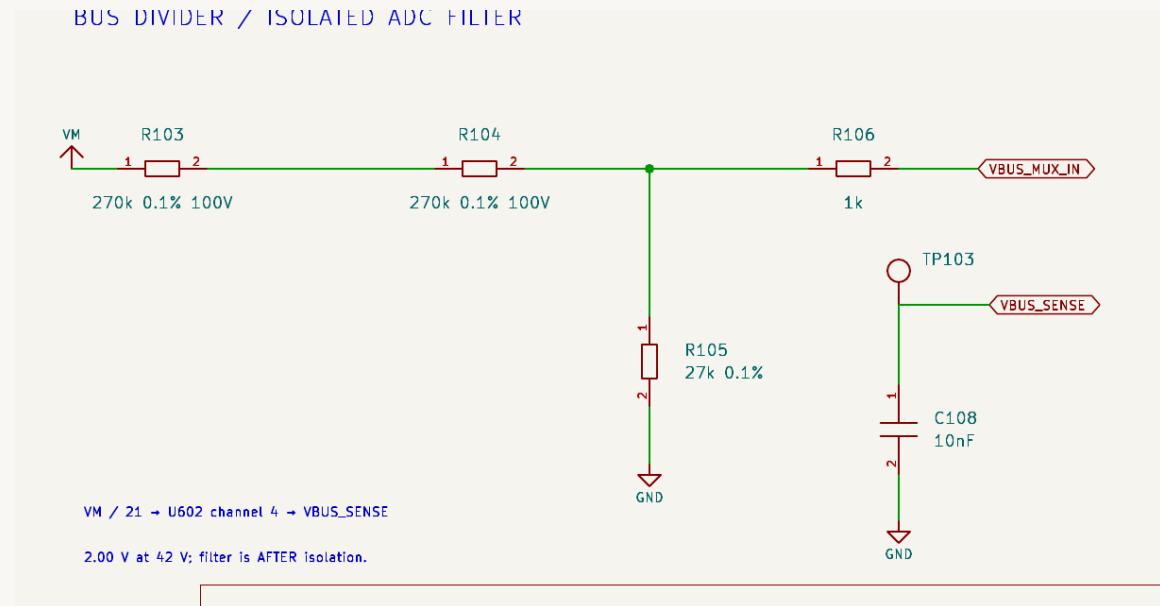
R101 dissipates only 3.75 mW at 42 V. Bulk-only discharge has an approximately 16-minute time constant.

Damping remains unpopulated

D101 and R102/C107 need cable and hot-plug measurements before selecting values.

$\tau = 470 \text{ k}\Omega \times 2040 \text{ }\mu\text{F} = 959 \text{ s}$; external precharge/current limiting is required for first tests.

Bus divider and powered-off ADC isolation



A 21:1 voltage scale

$R103 + R104 = 540 \text{ k}\Omega$. $R105 = 27 \text{ k}\Omega$.
Splitting the upper leg distributes voltage stress.

267 μs nominal filtering

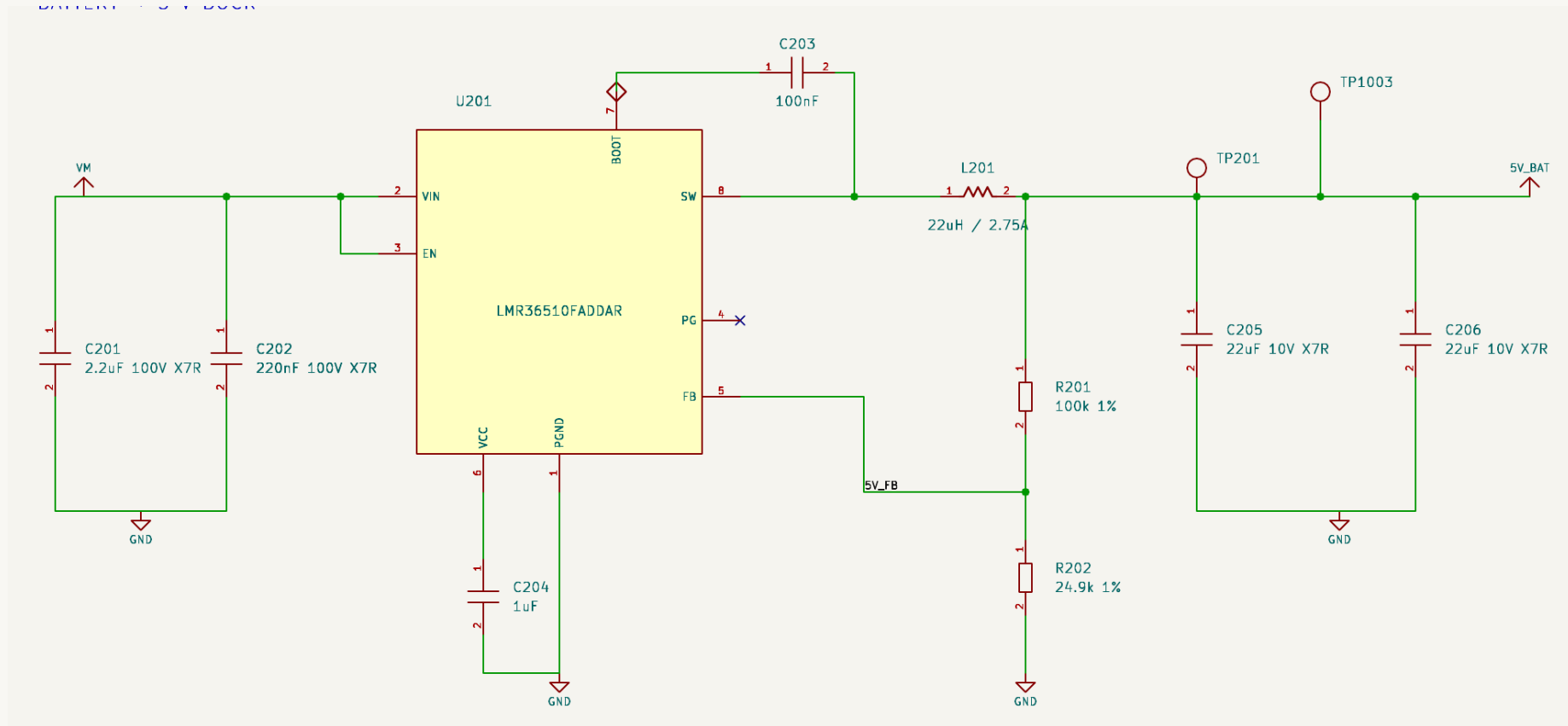
$R106 = 1 \text{ k}\Omega$ and $C108 = 10 \text{ nF}$. Divider impedance contributes most of the time constant.

U602 isolates the live divider

The capacitor sits after isolation. Enable the mux and wait at least 2 ms before trusting VM.

$V_{ADC} = VM / 21 = 2.00 \text{ V at } 42 \text{ V}$; VM sensing is invalid whenever the feedback mux is disabled.

Battery-to-5 V buck converter



LMR36510F at 400 kHz

A switching converter avoids dissipating the large VM-to-5 V drop as heat. Forced PWM keeps predictable switching at light load.

22 µH inductor

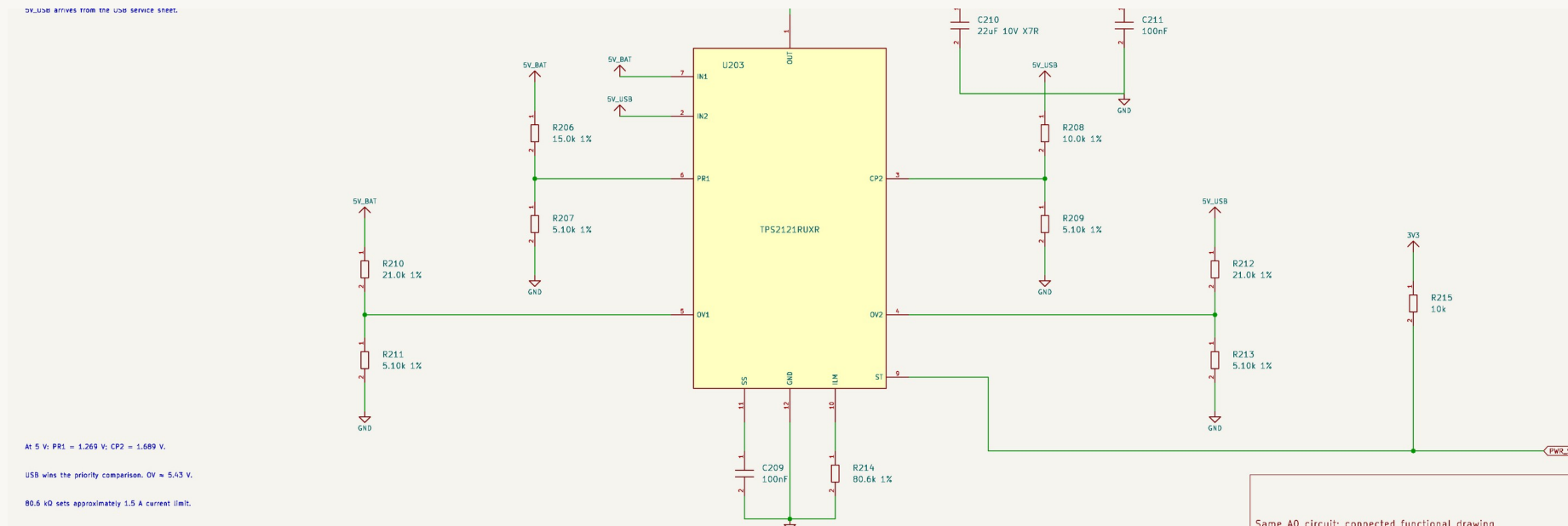
Nominal ripple is 0.41–0.50 A peak-to-peak over 18–42 V for approximately 5 V output.

44 µF nominal output bank

Two 22 µF ceramics support load transients. Effective capacitance and regulator stability still require review.

$$V_{OUT} = 1.00 \times (1 + 100 \text{ k}\Omega / 24.9 \text{ k}\Omega) = 5.016 \text{ V}$$

USB-priority power mux



At 5 V: PR1 = 1.269 V and CP2 = 1.689 V, so valid USB takes priority.

Different priority divider ratios

15 k Ω / 5.10 k Ω on battery and 10 k Ω / 5.10 k Ω on USB set USB preference.

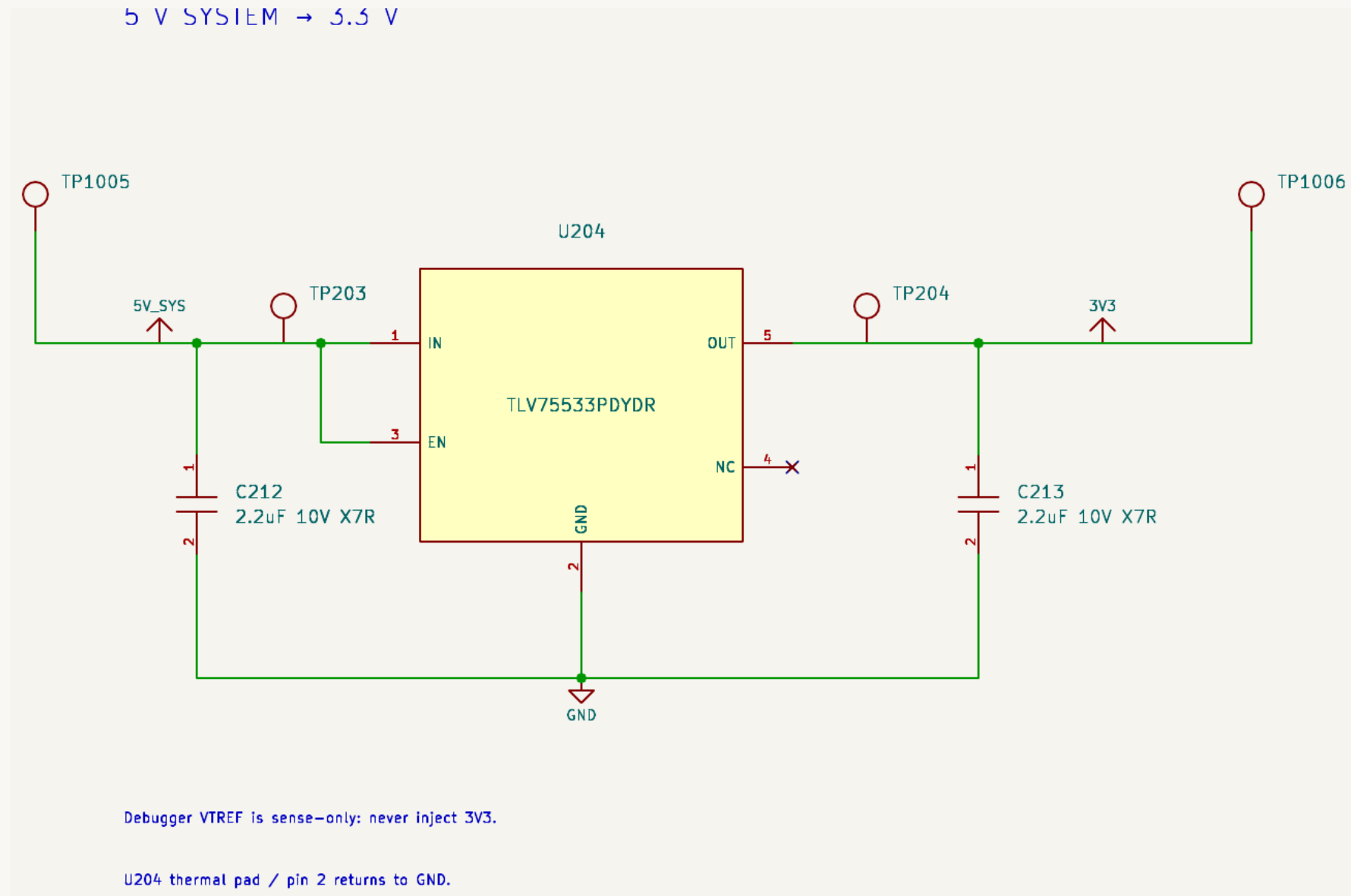
Defined input thresholds

21 k Ω / 5.10 k Ω dividers set approximately 5.43 V overvoltage thresholds.

Current limit and soft start

80.6 k Ω sets approximately 1.5 A. C209 = 100 nF controls startup slew.

3.3 V LDO and control-power budget



TLV75533P supplies the logic rail

The external 500 mA regulator supports both battery and USB operation while bypassing the STSPIN internal 3.3 V regulator.

250 mA planning allowance

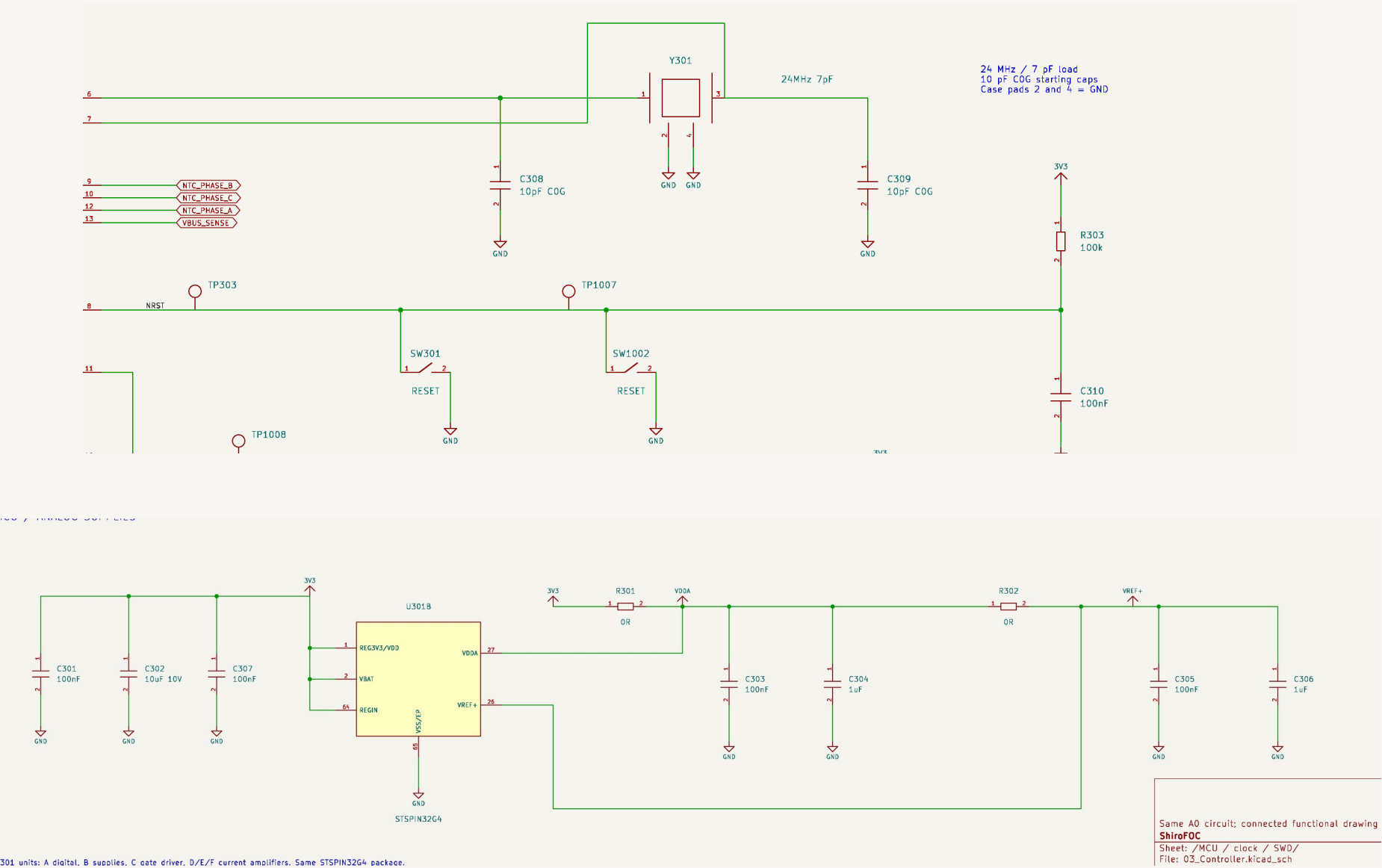
The recorded CAN-fault allowance raises this to 320 mA. External feedback is limited to 25 mA.

Thermal loss matters

At 5.25 V input, the LDO dissipates 0.488 W normally or 0.624 W at the fault allowance.

$$P_{LDO} = (V_{IN} - 3.3 \text{ V}) \times I_{LOAD}$$

MCU, clock, reset and SWD



24 MHz crystal, 7 pF load

Two 10 pF C0G capacitors contribute 5 pF series load, leaving approximately 2 pF for stray capacitance.

Controlled reset and boot states

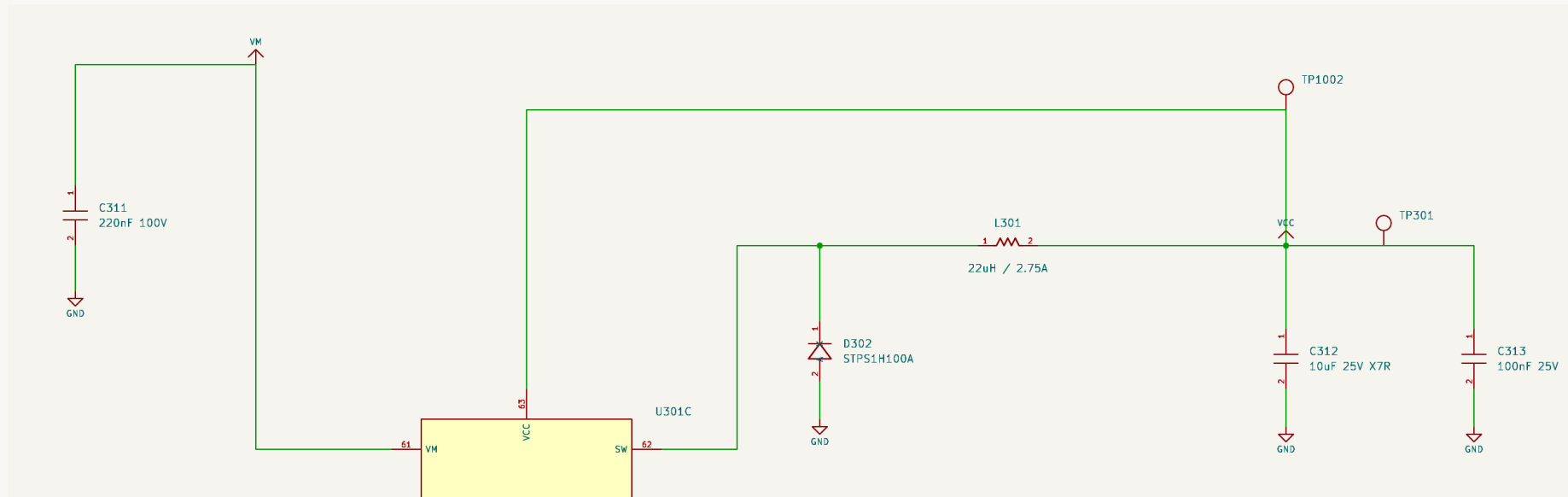
Reset pull-up/filter and BOOT0 defaults support predictable startup. PB8 shares the feedback index/Hall input.

Debug and analog supplies

SWD remains available. Local bypassing serves VDD, VDDA and VREF+. VTREF is a sense connection.

$$CL \approx C1C2 / (C1 + C2) + C_{stray}$$

Gate-driver supply and 10 V selection



8 V at reset, 10 V in operation

Firmware must restore the 10 V setting after every VM power-up.

22 μ H is a prototype substitution

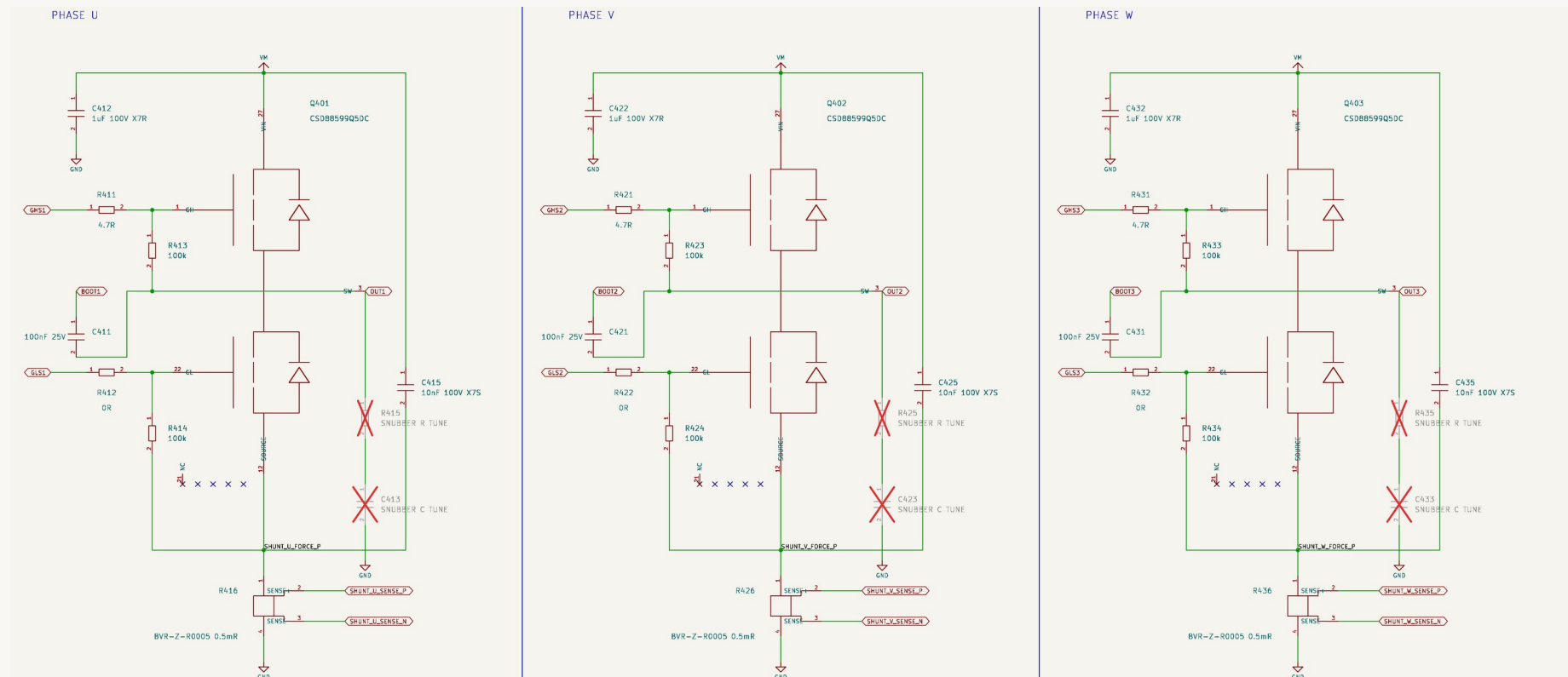
L301 replaces the typical 18 μ H example. Regulation and startup require measurement.

10 μ F reservoir ♦ 100 nF bypass

C312/C313 supply fast gate and bootstrap demand locally. Brake-driver demand also loads VCC.

$6 \times 56 \text{ nC} \times 20 \text{ kHz} = 6.72 \text{ mA}$ average gate-charge current

Three-phase inverter topology



One dual MOSFET package per phase

Q401–Q403 each contain the high-side and low-side switches. Parallel packages are deferred.

Complementary switching

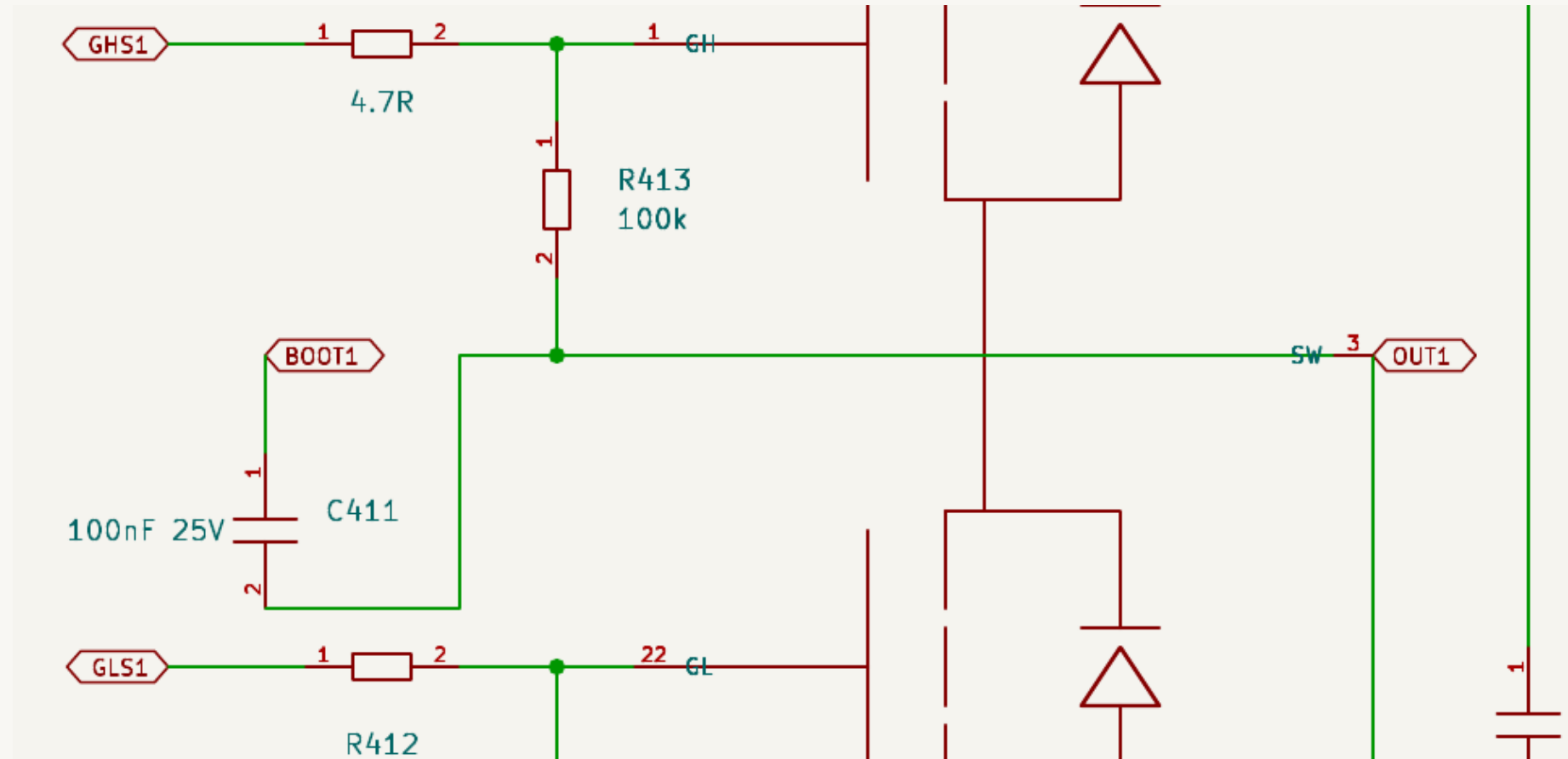
The upper switch connects a phase to VM.
The lower switch provides the lower rail
path and synchronous freewheeling.

Selection follows total loss

Conduction, switching, recovery and package thermal paths all matter alongside the 60 V rating.

At VGS = 10 V, datasheet RDS(on) is 1.7 mΩ typical / 2.1 mΩ maximum at its stated conditions.

Bootstrap operation and capacitor value



C411 / C421 / C431 = 100 nF, 25 V

Each capacitor floats between BOOTx and OUTx. It charges while the phase is low.

Gate-charge-only droop is 0.56 V

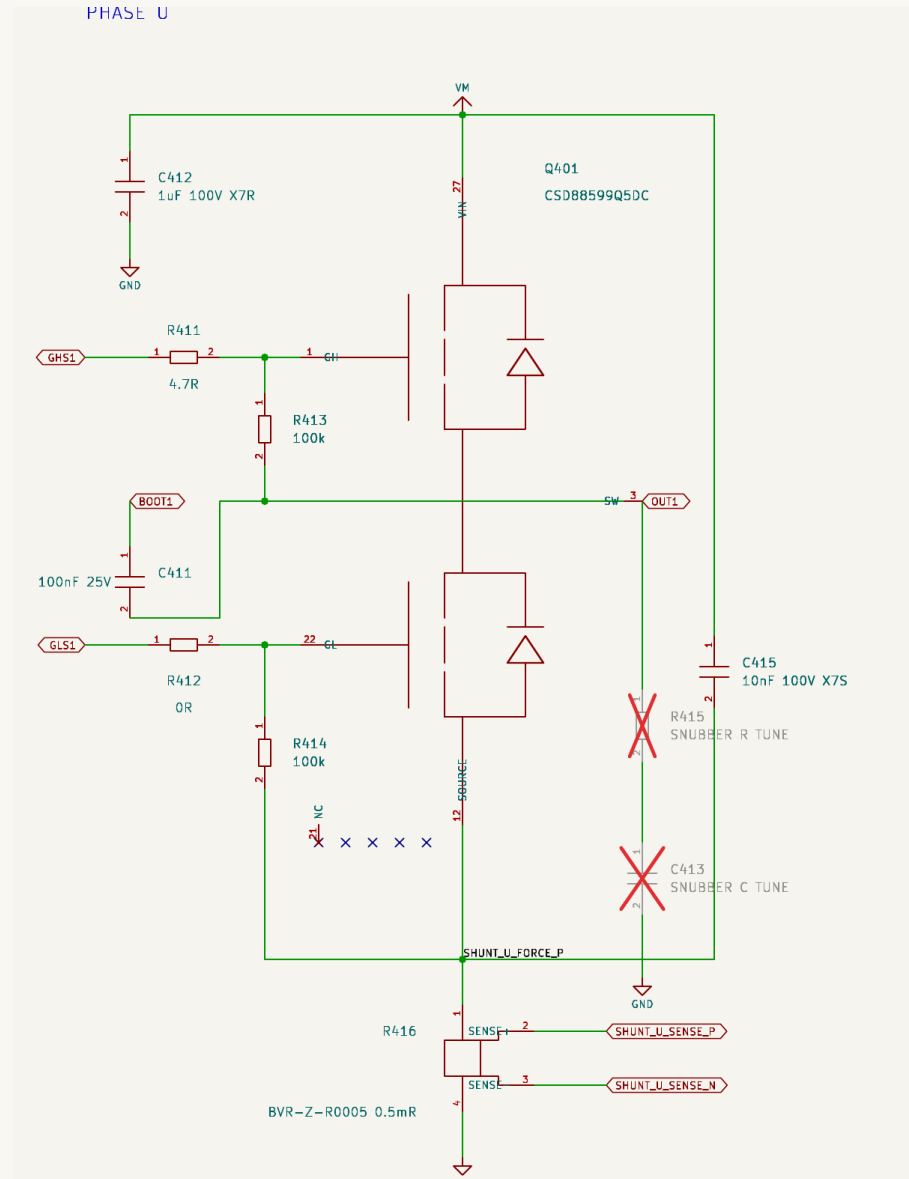
56 nC / 100 nF. Leakage, driver supply current and the 100 kΩ gate pull-down add charge demand.

Use effective capacitance

A 50 nF effective value would double that droop to 1.12 V. Larger values also take longer to recharge.

$$C_{eff} \geq [Q_g + (I_{HB} + I_{leak} + V_{GS}/R_{GS})t_{ON} + Q_{other}] / \Delta V_{allowed}$$

Bootstrap refresh and PWM limits



20 kHz gives a 50 μ s period

Low-side time must support both bootstrap replenishment and valid current sampling.

Internal charging resistance matters

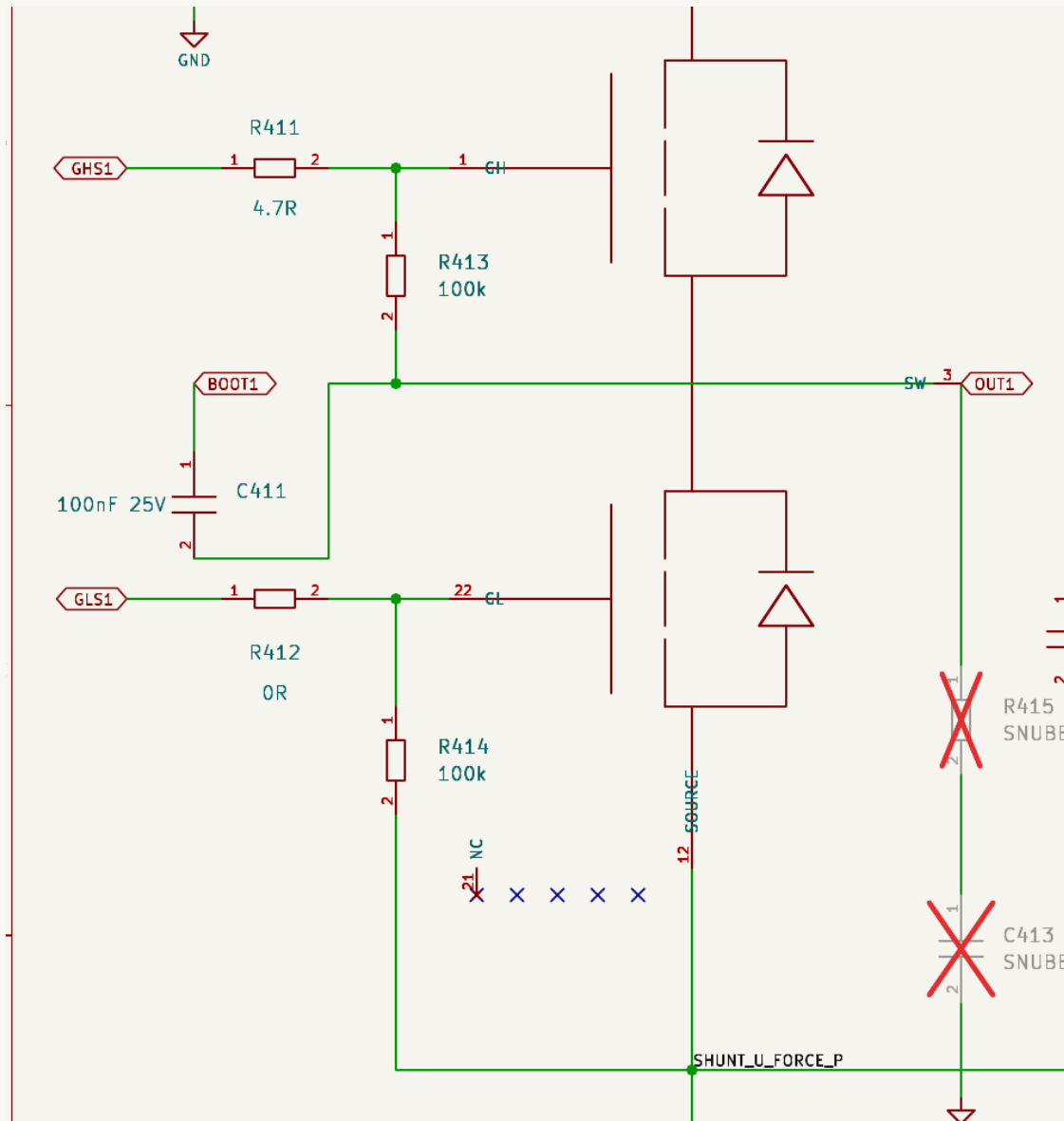
The local ST datasheet lists 100 Ω typical, 200 Ω maximum. With 100 nF, the simple RC scale is 10–20 μ s.

No static high-side 100% duty

Precharge at startup and enforce refresh intervals. Determine the modulation limit from measured BOOT–OUT voltage.

An RC illustration: 90% recharge from empty takes about $2.3\tau = 23\text{--}46 \mu\text{s}$.

High-side and low-side gate resistor choices



4.7 Ω at GH, 0 Ω at GL

R411/R421/R431 slow the high-side edge. R412/R422/R432 retain a strong low-side gate connection.

The asymmetry follows TI guidance

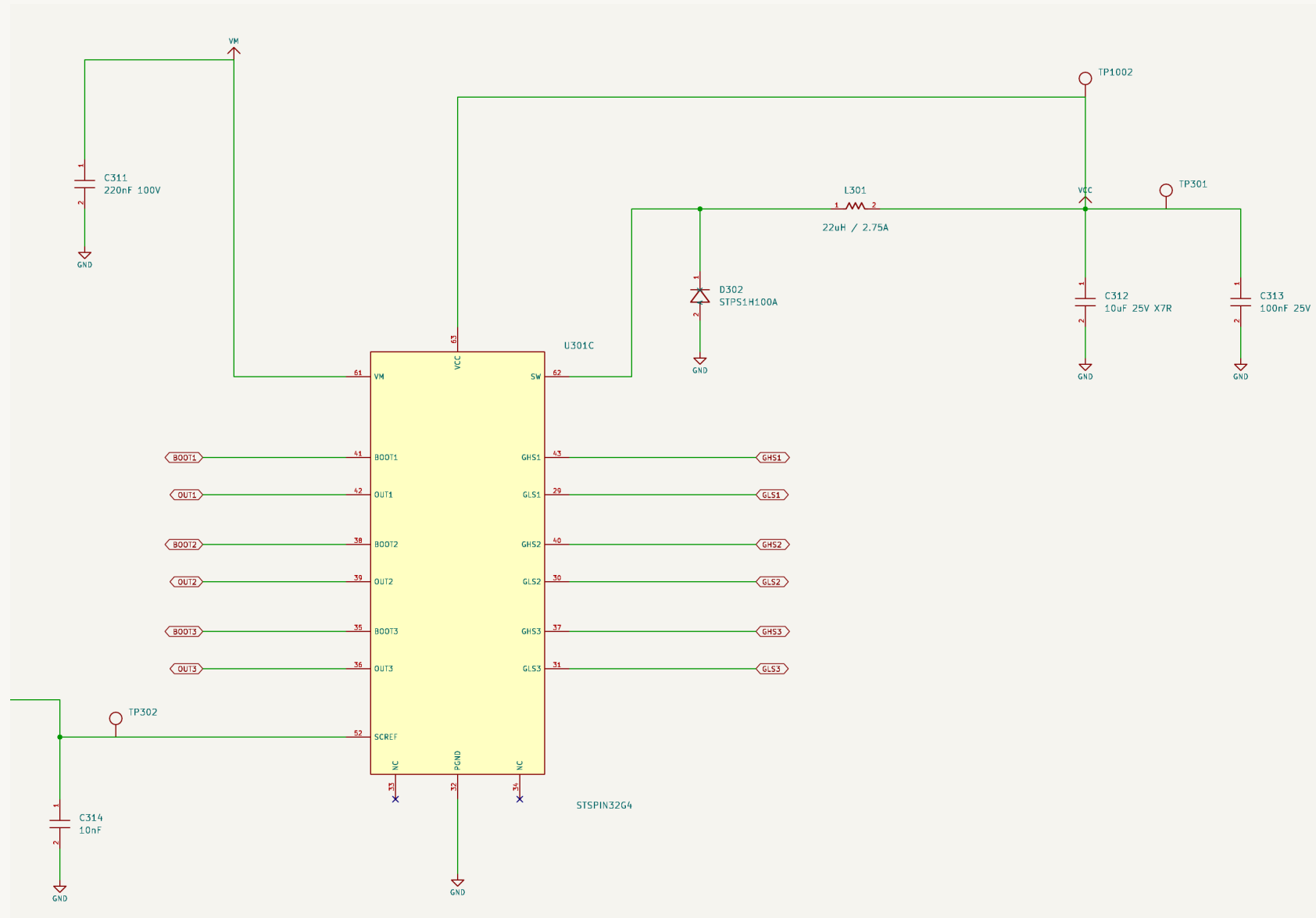
TI recommends 4.7–10 Ω on GH and a direct GL connection to reduce parasitic capacitive turn-on.

Tune with the actual driver and PCB

Larger resistance can reduce ringing but increases switching loss. A 100 k Ω pull-down establishes the unpowered default.

$$I_{g,Miller} \approx (V_{drive} - V_{plateau}) / (R_{driver} + R_{external} + R_{gate}); t_{Miller} \approx Q_{gd} / I_g$$

Deadtime, interlocking and switching losses



Start at 500 ns deadline

The MCU begins with conservative timing at 20 kHz. Driver interlocking provides a separate layer.

Longer deadtime also costs energy

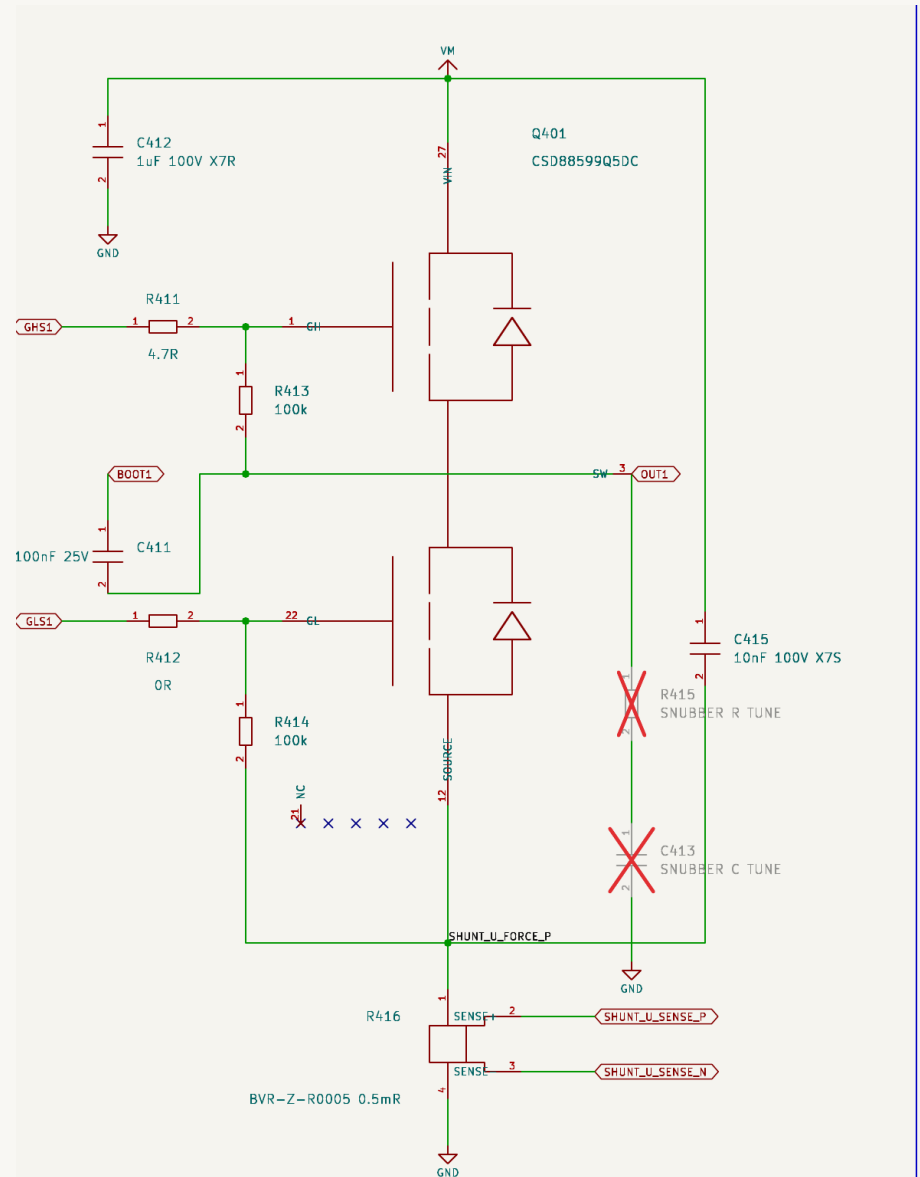
Inductive current uses body diodes during the interval, affecting loss and low-current distortion.

Gate speed is a measured tradeoff

Propagation delay, Miller behavior and temperature determine the minimum safe setting.

$$P_{\text{cond}} \approx I_{\text{rms}}^2 R_{\text{DS(on),T}}; P_{\text{sw}} \approx \frac{1}{2} V_{\text{DSID}} (t_r + t_f) f_{\text{sw}}$$

Local capacitors, ringing and snubbers



Two different return points

C412 = 1 μ F to GND. C415 = 10 nF directly to the source above the shunt.

R415 / C413 are DNP

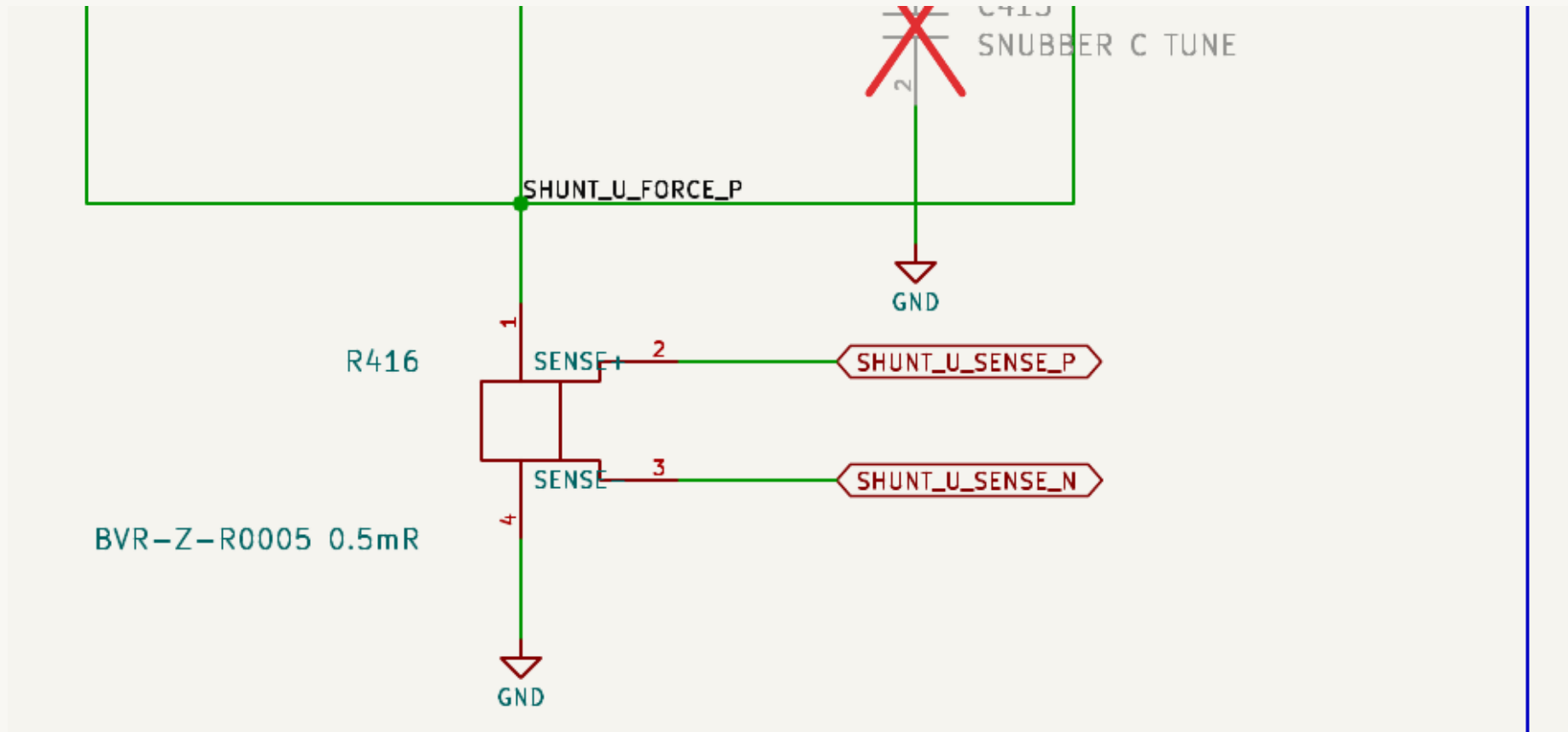
A series RC branch can damp phase-node resonance after measurement identifies the ringing.

42 V and regeneration need attention

TI calls for RC snubbers in its 42–54 V input region. This board has no fitted phase snubber yet.

Commissioning target: measured VDS peaks below 55 V, with margin to the 60 V absolute limit.

Three-shunt current sensing



0.5 mΩ per phase

R416/R426/R436 produce 20 mV at 40 A and 40 mV at 80 A.

Kelvin pads measure the element

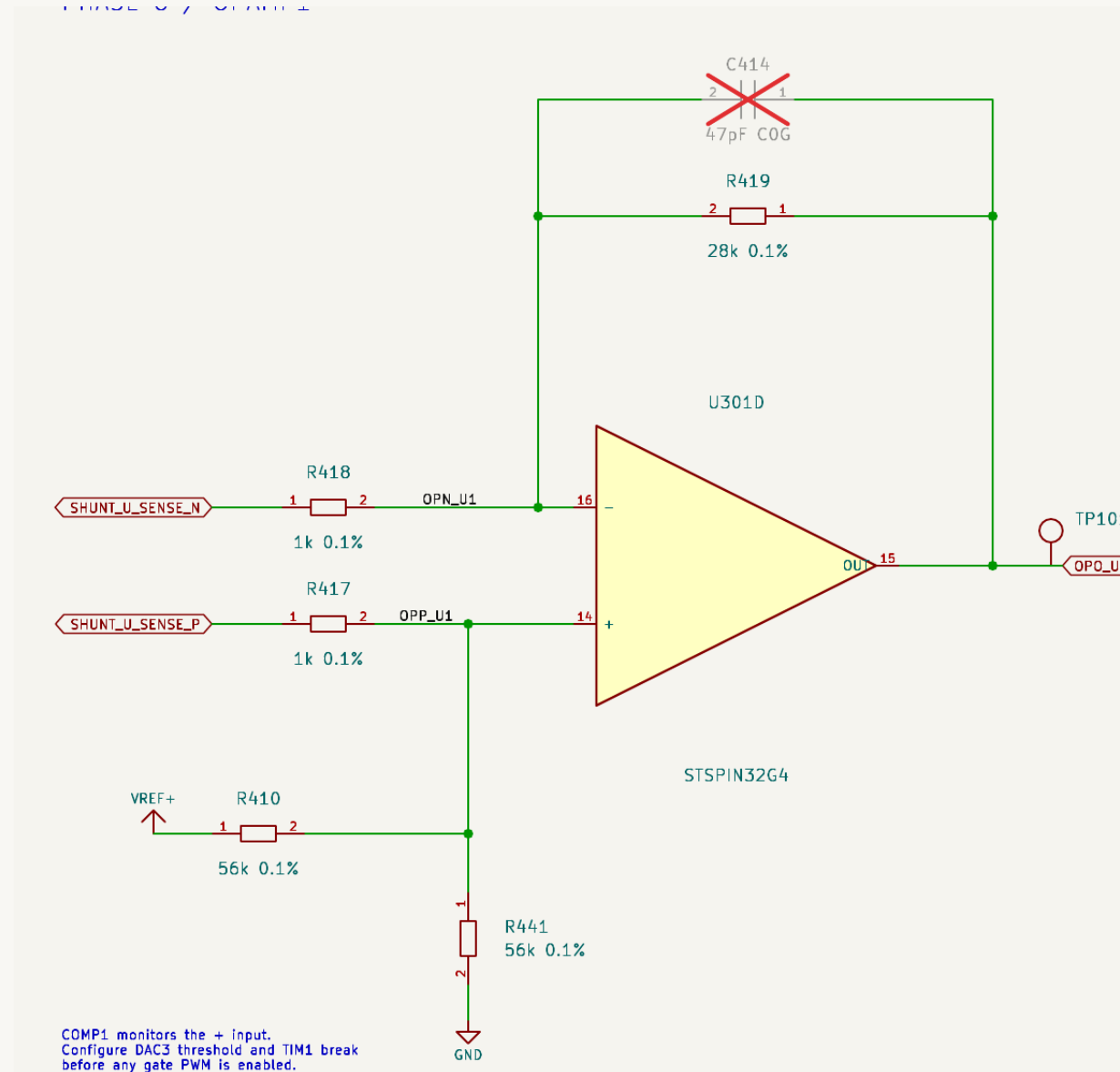
Force current uses pads 1/4. Sense pads 2/3 connect only to the amplifier network.

Signal size versus loss

Lower resistance saves power but makes offsets and noise more significant. Three shunts improve sensing options.

$P_{\text{shunt}} = I^2 R = 0.8 \text{ W at } 40 \text{ Arms; } 3.2 \text{ W instantaneously at } 80 \text{ A}$

Current-amplifier gain, bias and tolerance



Gain 28, centered at 1.65 V

1 kΩ input resistors and 28 kΩ feedback provide 14 mV/A with the 0.5 mΩ shunt.

Matched 0.1% resistor network

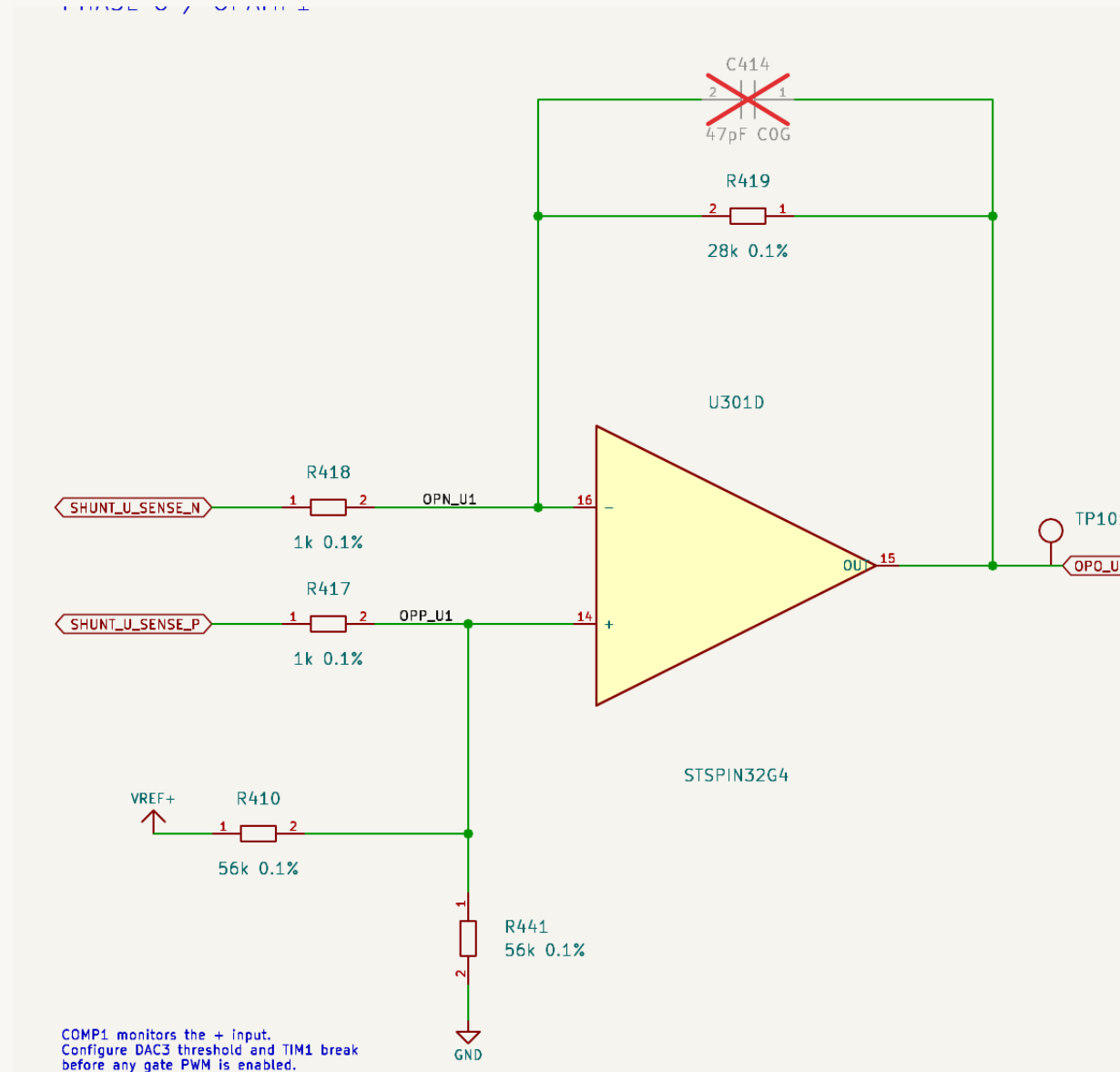
Two 56 kΩ resistors form an equivalent 28 kΩ bias source at VREF/2.

Bipolar measurement with headroom

−80 A, zero and +80 A correspond nominally to 0.53 V, 1.65 V and 2.77 V.

$$V_{OUT} = V_{REF} / 2 + 28(V_{SENSE+} - V_{SENSE-})$$

Sampling and hardware overcurrent shutdown



Comparators see the input node

COMP1/2/4 observe the biased op-amp positive inputs, before amplification.

♦60 A is an initial trip target

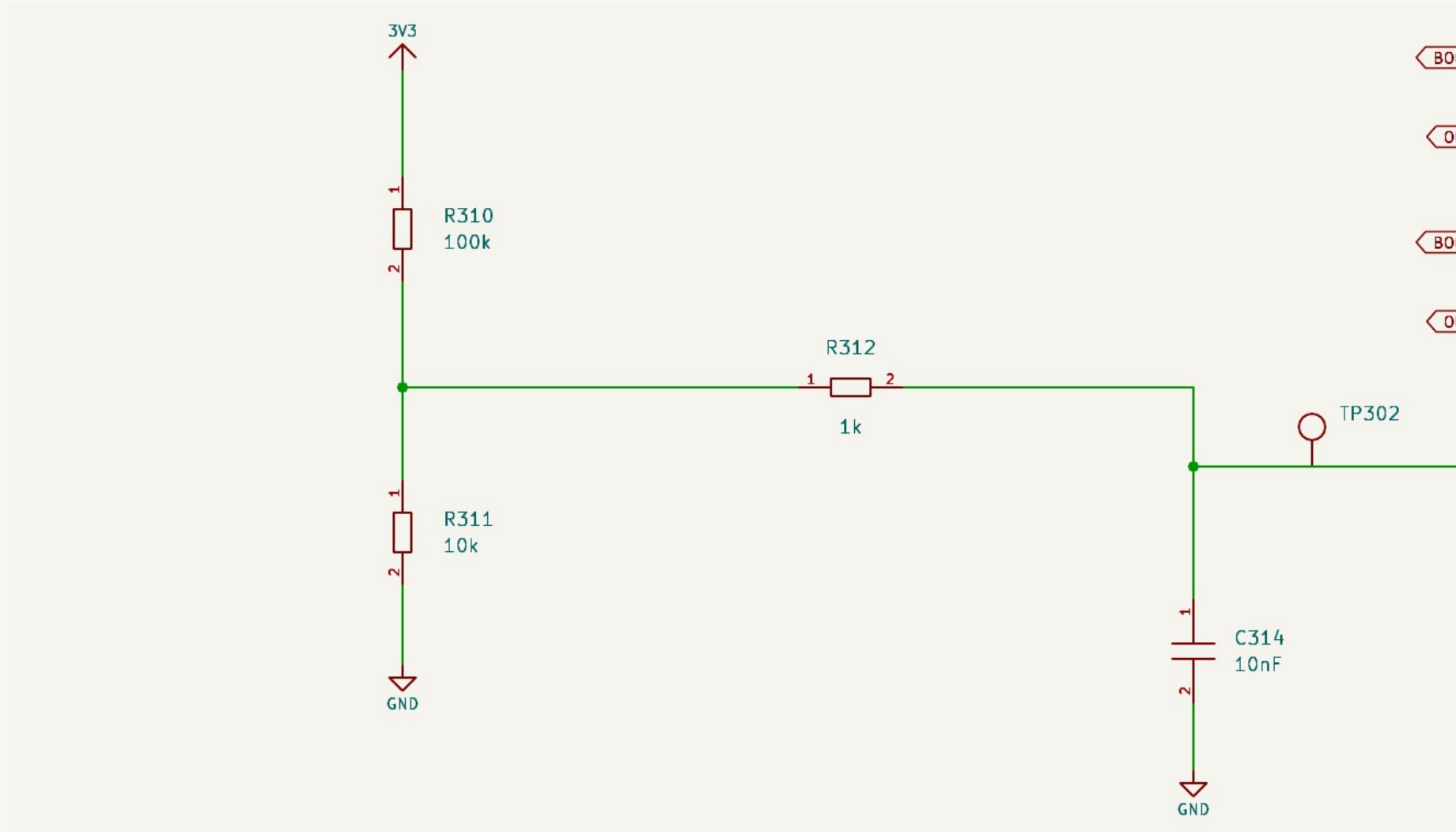
The nominal input is 85.86 mV, about DAC code 107. Offset and DAC errors can move the actual trip substantially.

Timer break acts after configuration

The hardware path can stop PWM without a CPU interrupt. Negative-current limits also need ADC supervision.

$$V_{COMP} \approx (56 \times 0.0005 \times I + 3.3) / 58$$

VDS fault detection and SCREF



0.30 V nominal reference

R310 = 100 k Ω and R311 = 10 k Ω divide the 3.3 V rail.

Local reference filtering

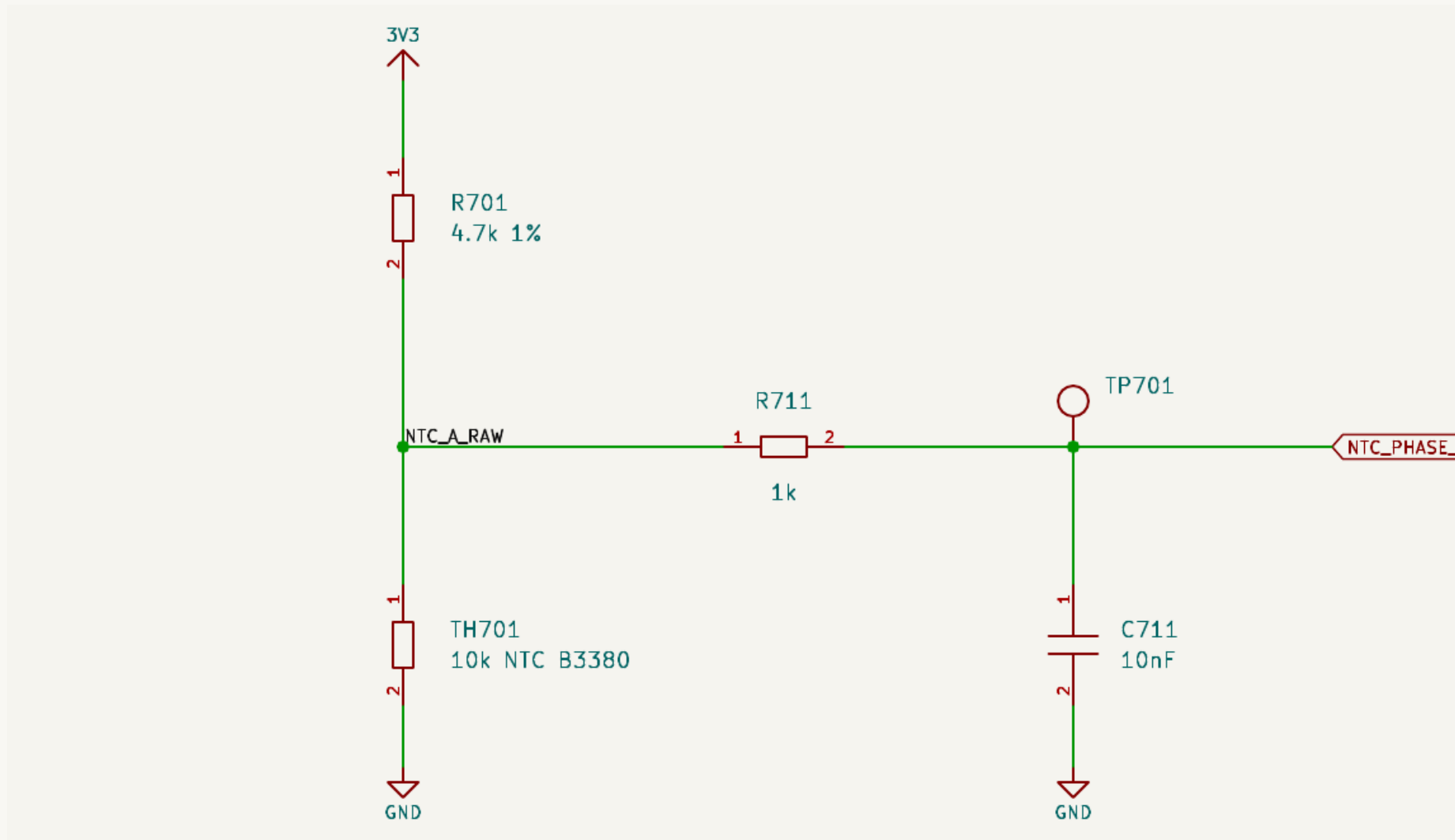
R312 = 1 k Ω and C314 = 10 nF filter SCREF. Divider impedance makes τ approximately 101 μ s.

An additional short-circuit mechanism

VDS depends on current, RDS(on), temperature and gate enhancement. It cannot set a precise phase-current limit.

$$V_{SCREF} = 3.3 \times 10 / 110 = 0.30 \text{ V}$$

Phase temperature measurement



Three 10 kΩ NTCs, B3380

TH701–TH703 monitor local phase heating. Each uses a 4.7 kΩ pull-up.

Sensitivity shifts toward warm operation

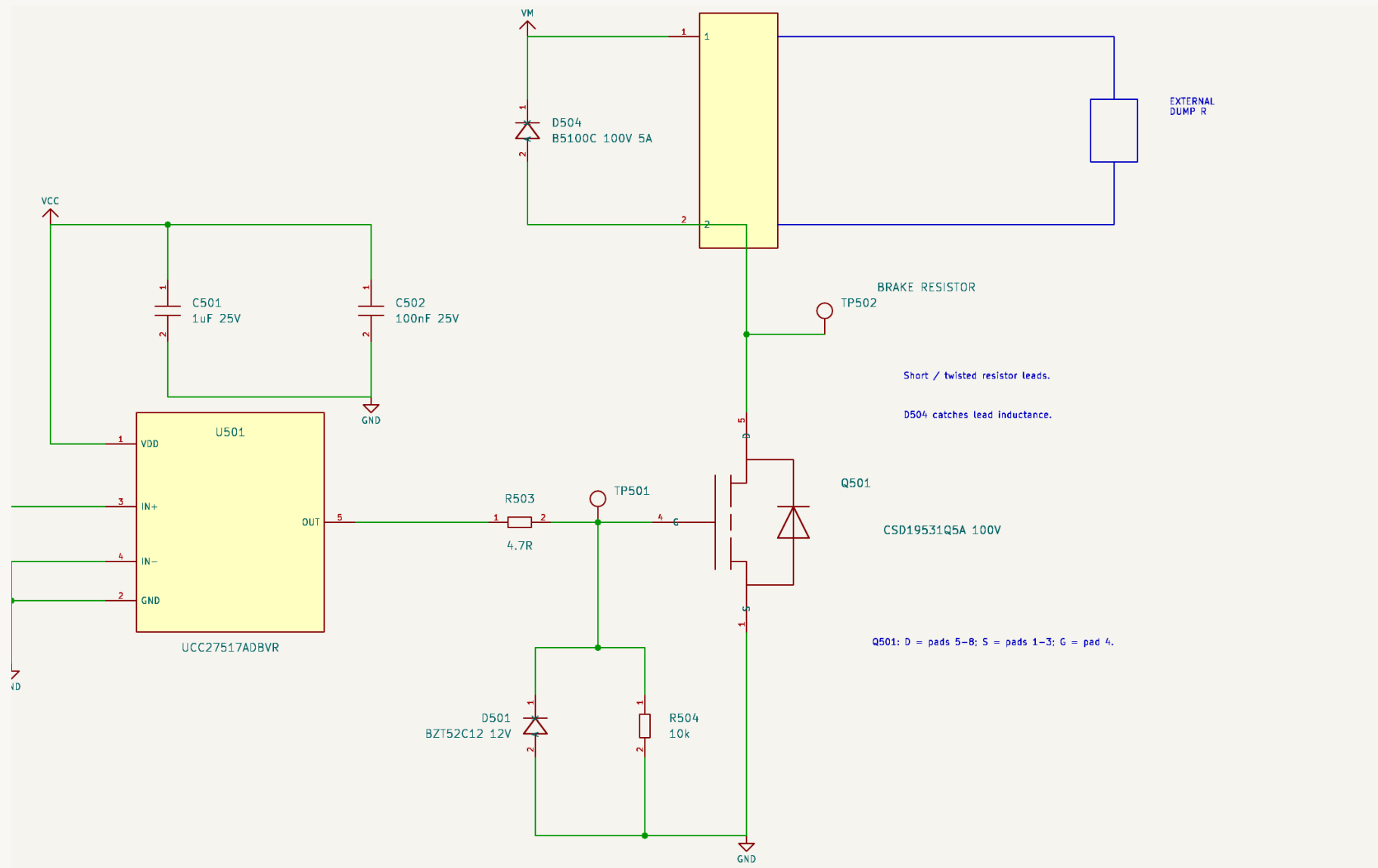
The lower fixed resistor improves the useful divider span at elevated temperature. Each ADC has 1 kΩ / 10 nF filtering.

Use the hottest valid sensor

An open NTC reads high, a short reads low. Sensor temperature lags MOSFET junction temperature.

$$V_{ADC} = 3.3 \times R_{NTC} / (4.7 \text{ k}\Omega + R_{NTC})$$

Brake-chopper topology and gate drive



A local energy dump

The external resistor connects from VM to BRK_SW. Q501 switches its return to ground.

100 V MOSFET and dedicated driver

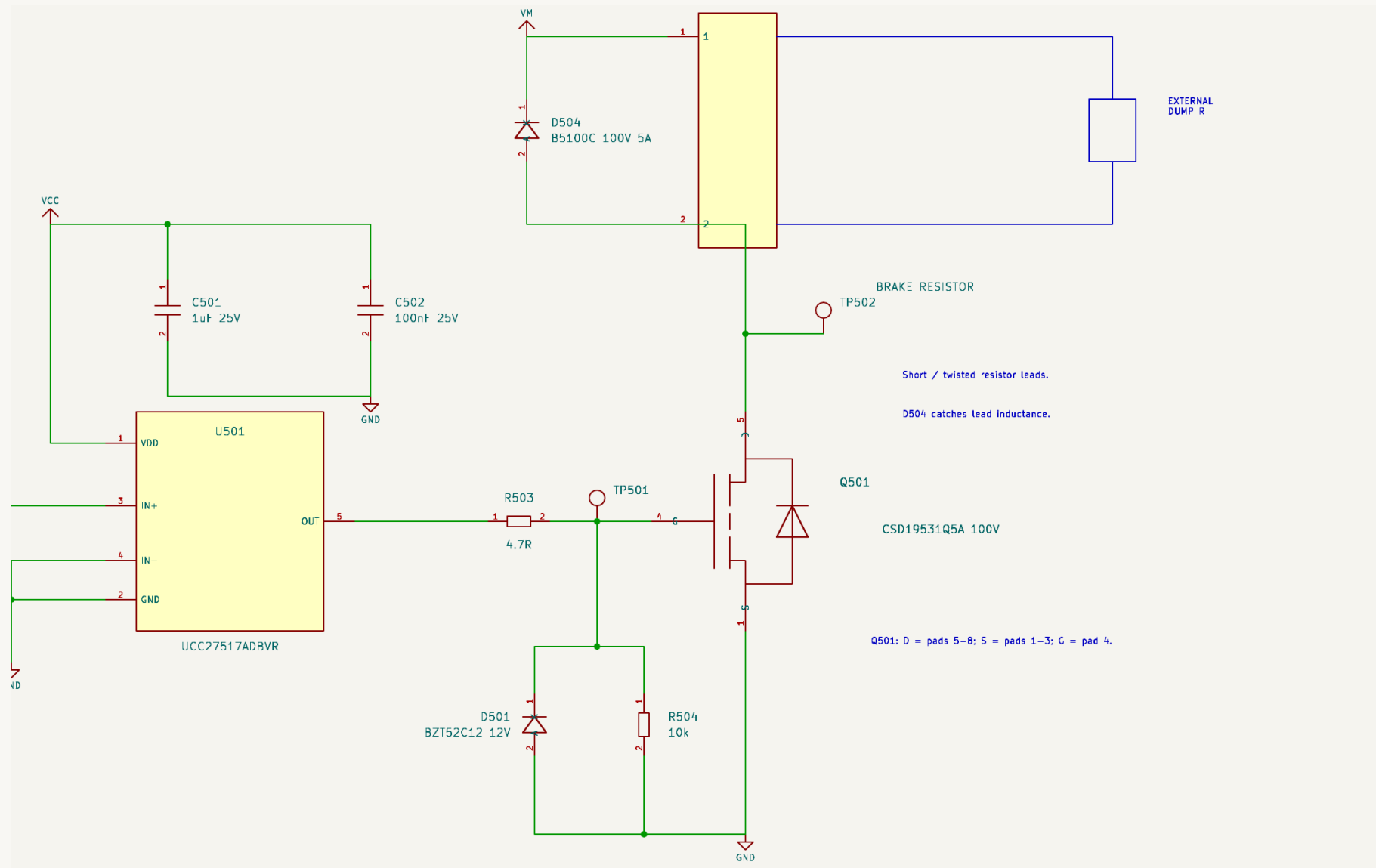
CSD19531Q5A provides added voltage margin. UCC27517A drives it from VCC.

A controlled gate loop

4.7 Ω gate resistor, 10 kΩ pull-down, 12 V gate clamp and local 1 μF + 100 nF bypass.

D504 recirculates brake-lead inductance. The external resistor absorbs sustained regenerative energy.

Brake resistor power and energy sizing



10 Ω at a 48 V calculation point

The resistor draws 4.8 A and dissipates 230.4 W when fully on.

A stop can contain 219 J

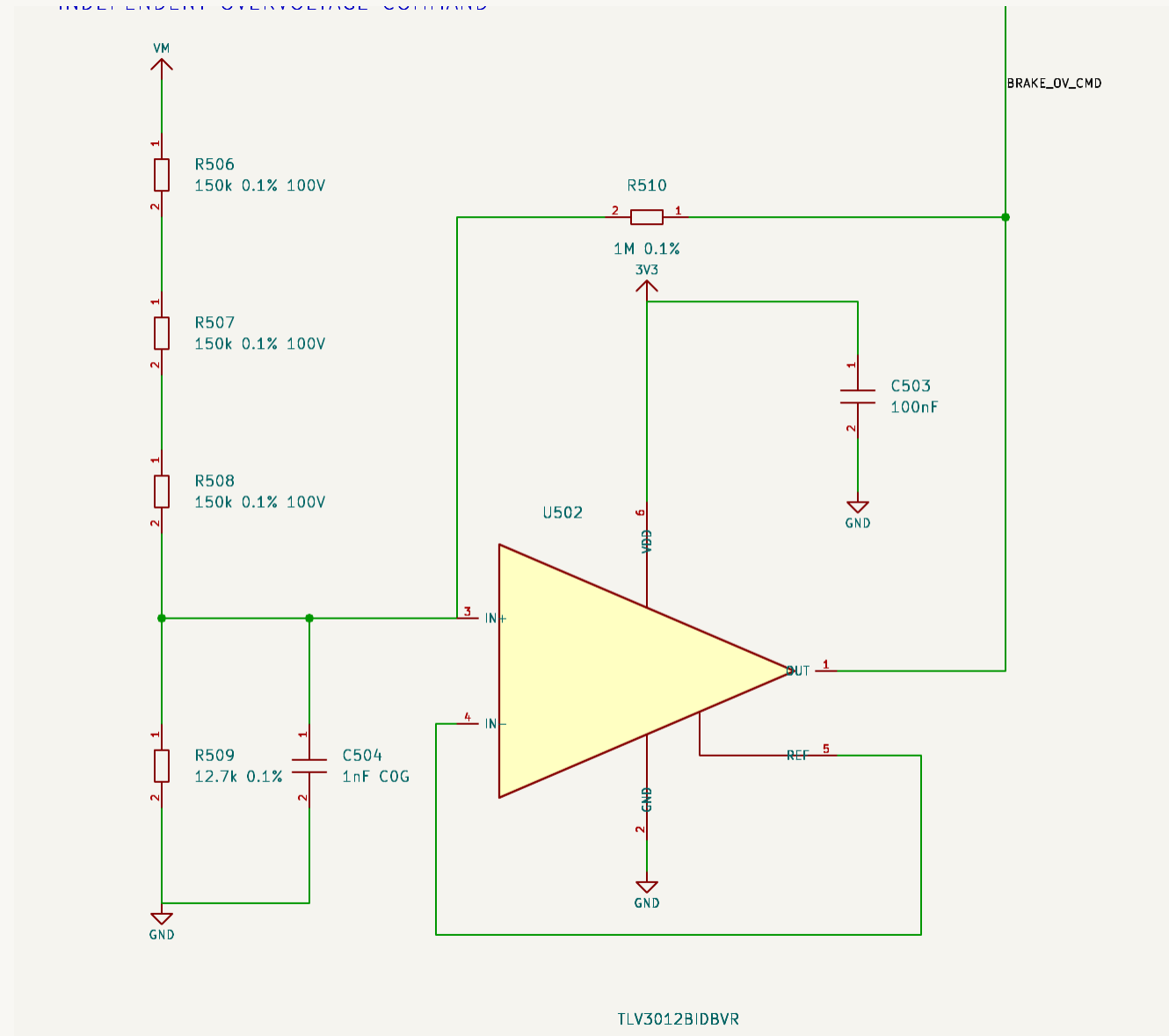
Example: $J = 0.01 \text{ kg}\cdot\text{m}^2$, speed = 2000 rpm, final speed = zero.

Power and pulse energy are separate

The example requires $\geq 250 \text{ W}$ with the specified heatsink and independently qualified 500 J over 5 s.

$$I = V/R; P = V^2/R; E = \frac{1}{2}J(\omega_{\text{start}}^2 - \omega_{\text{end}}^2)$$

Independent overvoltage brake command



TLV3012B with positive feedback

Three 150 k Ω upper resistors, 12.7 k Ω lower resistor and 1 M Ω hysteresis feedback.

45.92 V on / 44.21 V off

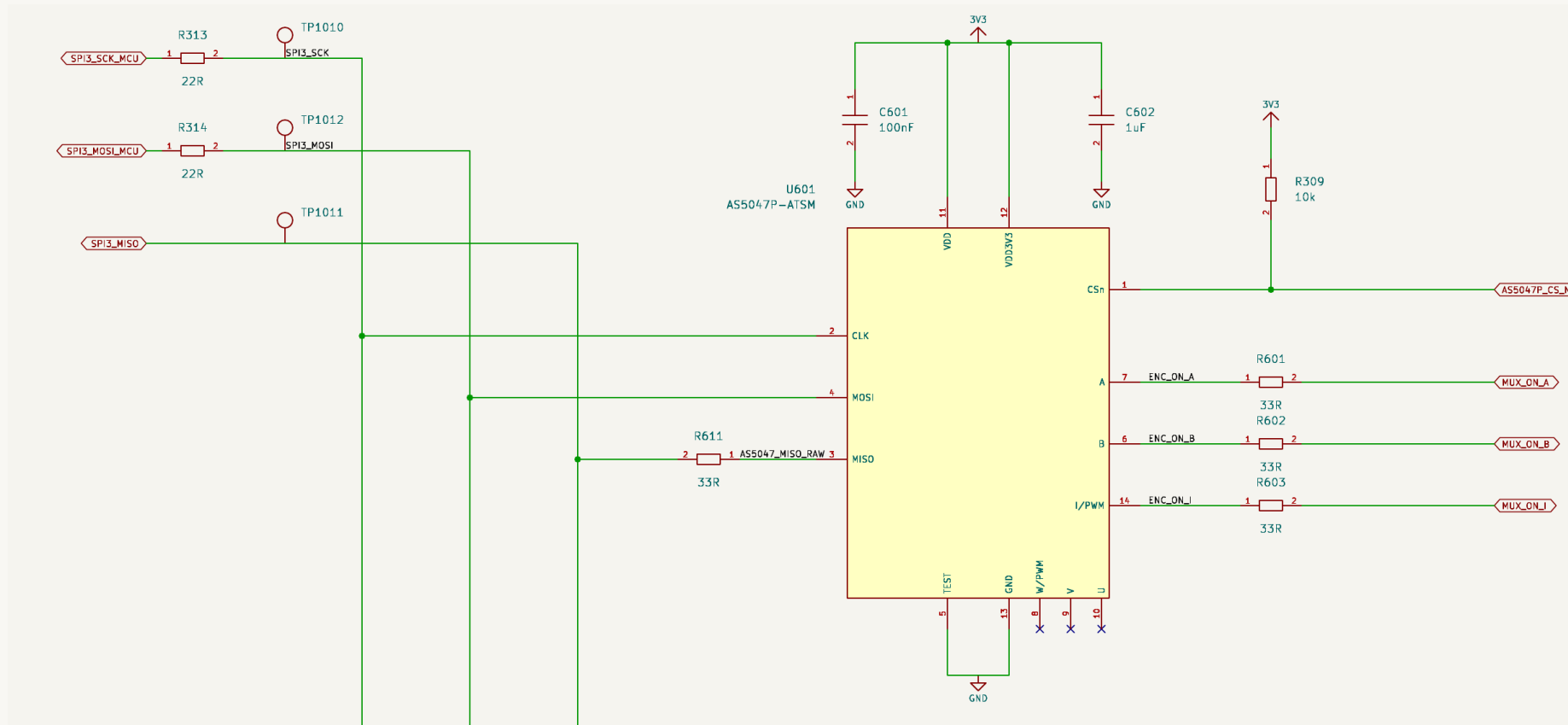
Recorded nominal thresholds. Corner ranges widen to 44.19–47.51 V rising and 42.58–45.96 V falling.

Firmware brakes earlier

43 V is the initial 10S firmware setting. The hardware OR path can act during MCU reset if 3V3 and VCC remain present.

The threshold calculation excludes wiring inductance and dynamic bus overshoot.

Onboard magnetic encoder



AS5047P position feedback

SPI provides absolute angle. ABI outputs provide timer-compatible incremental feedback through the selector.

Local supply and edge control

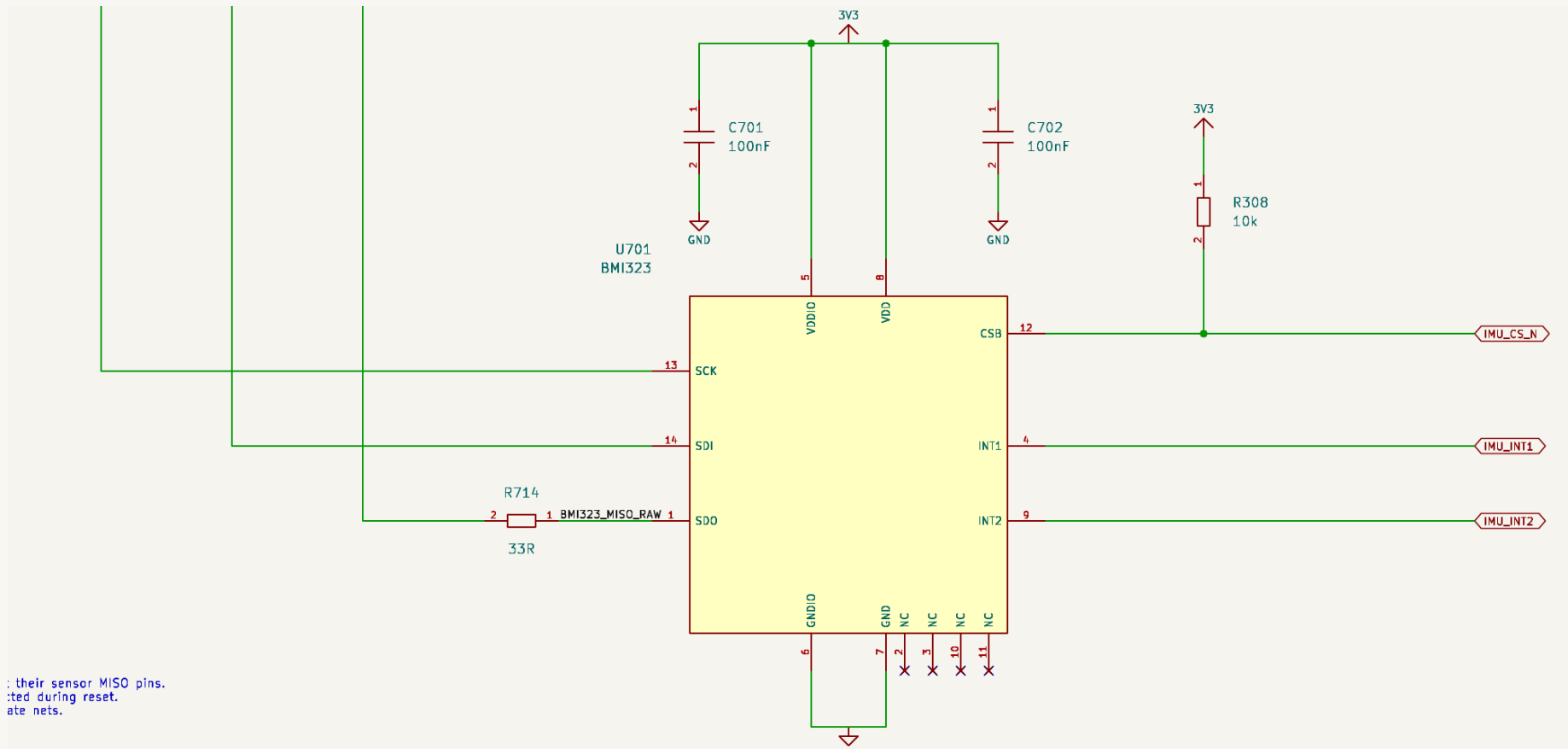
100 nF + 1 μ F bypassing. 33 Ω ABI and MISO series resistors are signal-integrity starting values.

Mechanical alignment is essential

Shaft magnet placement, offset calibration and error checks determine usable angle accuracy.

One sensor supports absolute SPI angle and incremental A/B/index paths.

IMU and shared SPI bus



BMI323 supplies motion telemetry

Its accelerometer/gyroscope data complements rotor feedback and uses separate interrupt pins.

A shared bus saves MCU pins

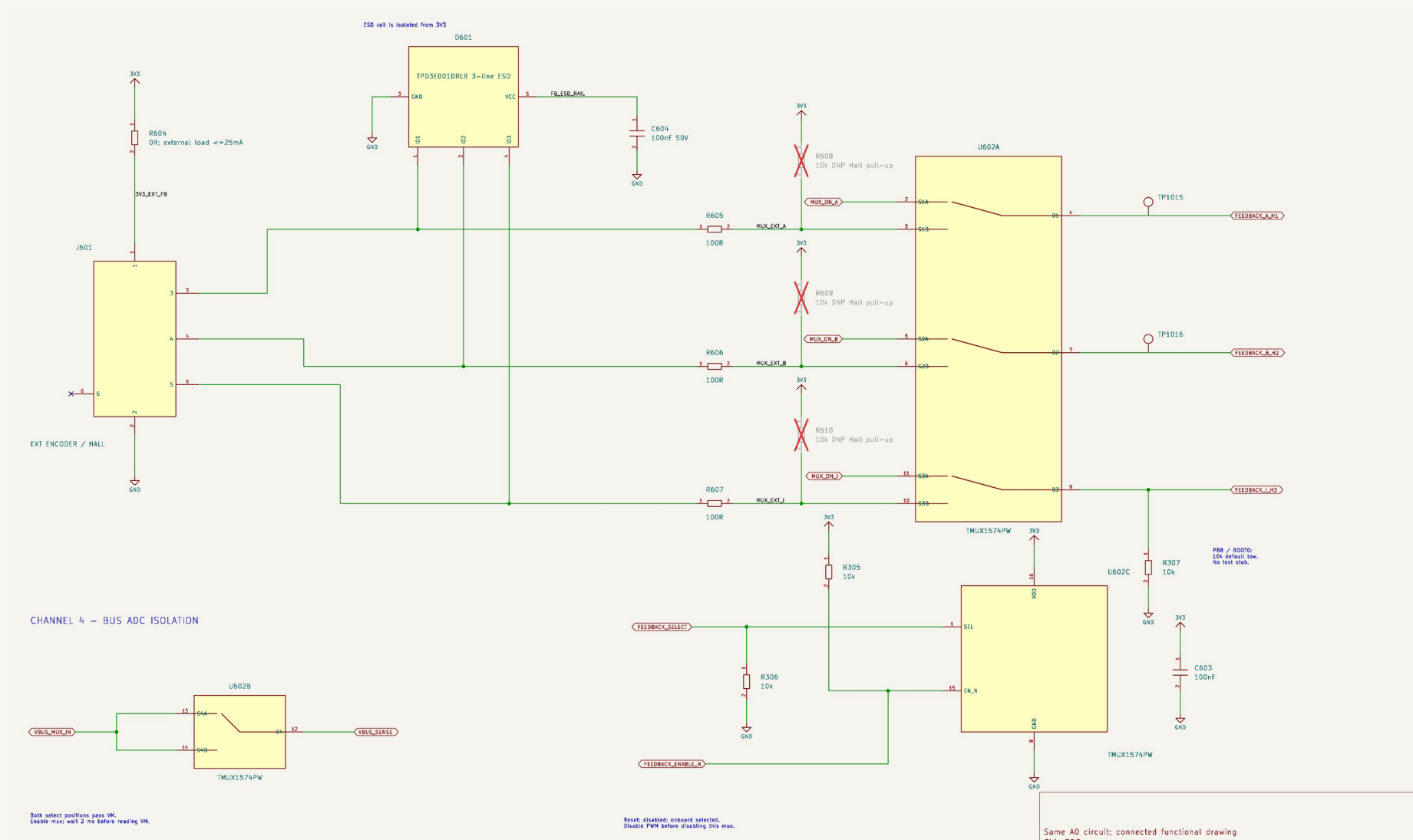
SCK/MOSI have 22 Ω source resistors.
Each peripheral MISO has its own 33 Ω series resistor.

SPI modes differ

Encoder mode 1, BMI323 mode 0 or 3.
Both chip selects must be high while the
MCU changes mode.

Keep sensor transactions separate and verify MISO release by the inactive device.

External feedback selection and protection



Onboard ABI or external Hall/encoder

TMUX1574 routes a selected source to the shared TIM4 inputs.

0–3.3 V signal contract

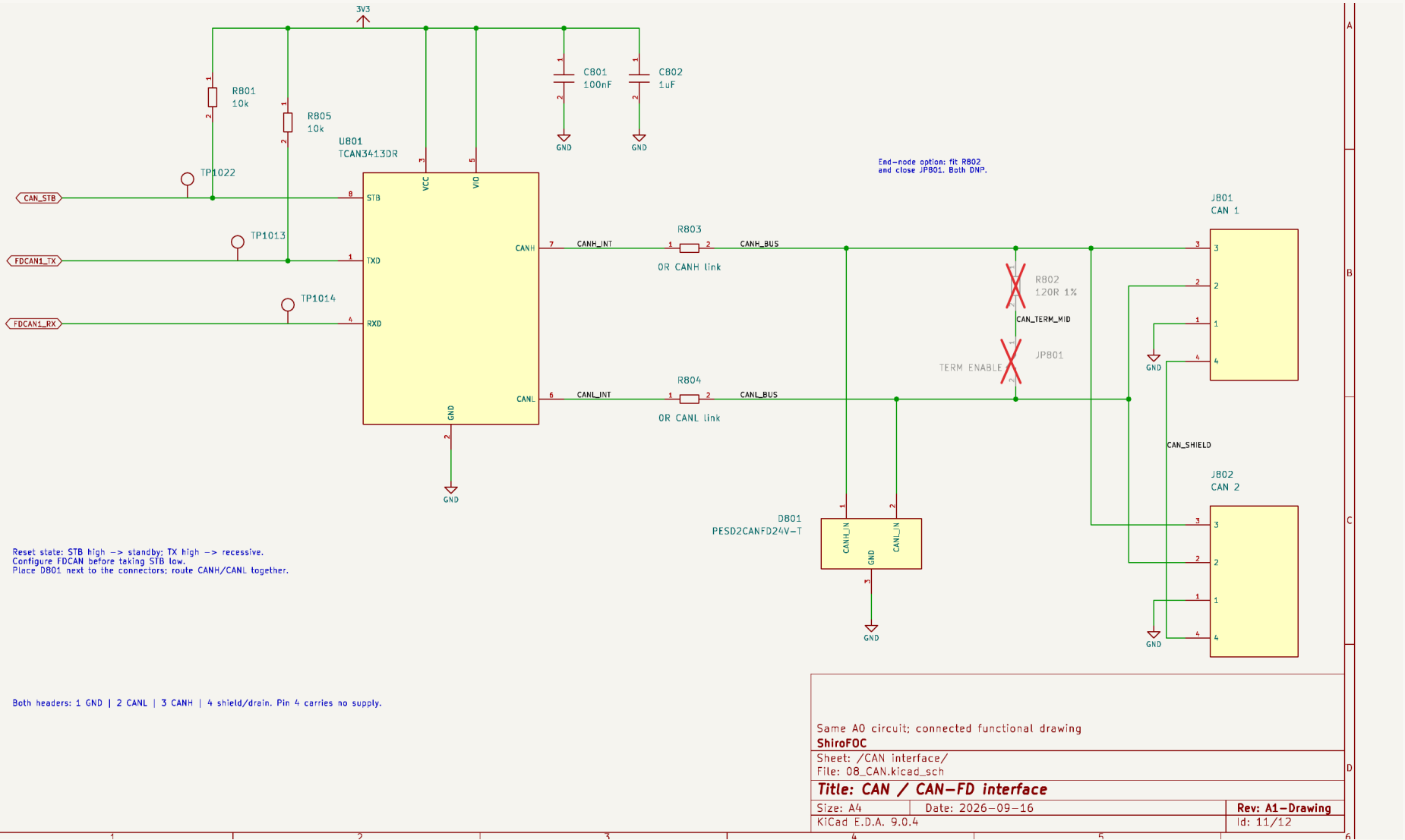
100 Ω input series resistors and ESD protection support the interface. Optional 10 k Ω pulls serve open-drain Hall outputs.

Reset isolation has two purposes

It protects PB8/BOOT0 state and disconnects live VM sensing from the unpowered MCU.

25 mA external supply allowance; R604 = 0 Ω is a link, not a current limiter.

CAN physical interface



TCAN3413 on the 3.3 V rail

The transceiver connects local FDCAN to a protected differential daisy-chain bus.

Standby and recessive defaults

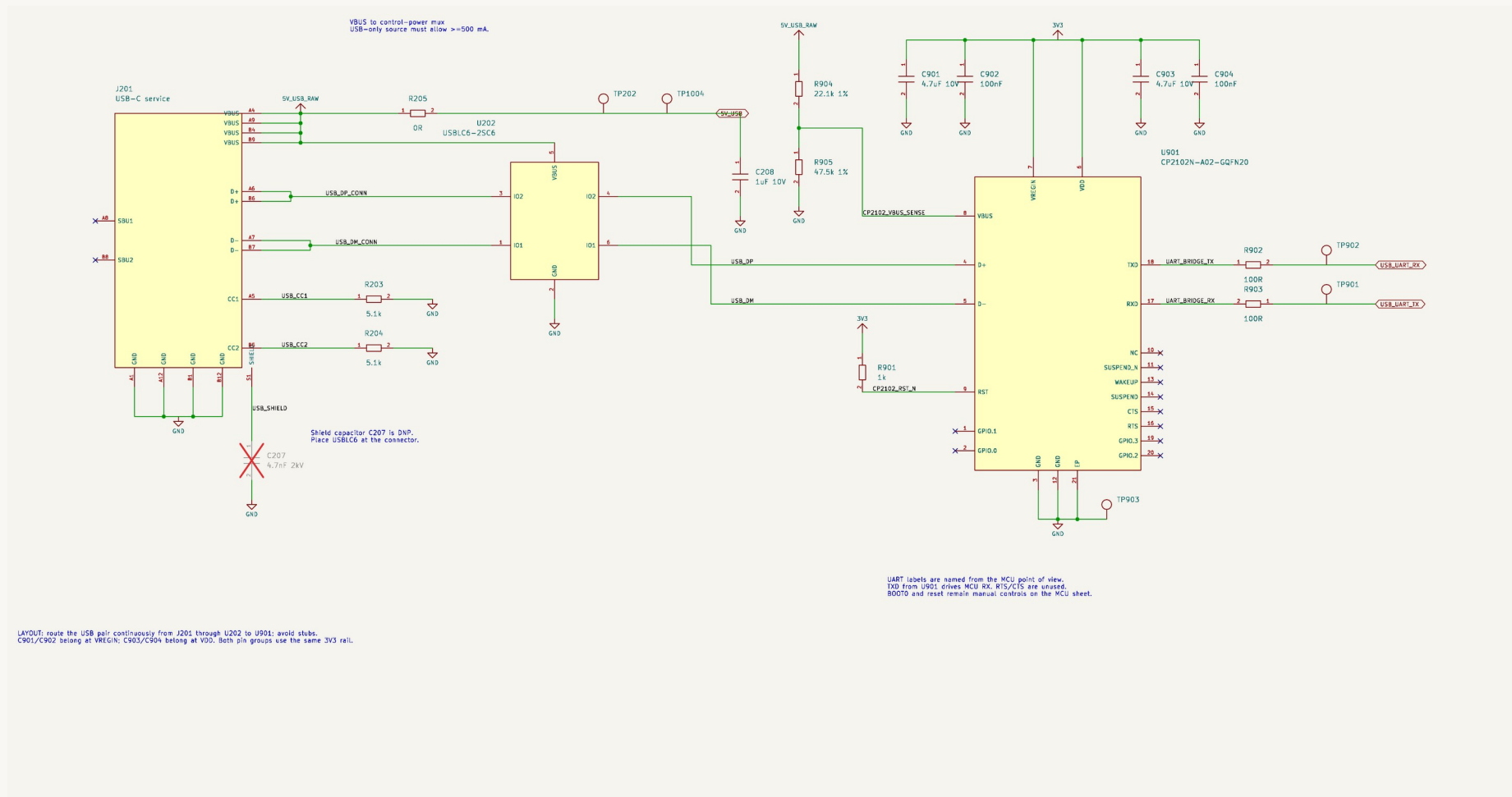
10 kΩ pulls hold STB and TX high until firmware configures the interface.

Optional 120 Ω end termination

R802 and JP801 are DNP. Fit and close only at a physical bus end.

J801/J802: 1 GND, 2 CANL, 3 CANH, 4 shield/drain. Pin 4 carries no supply.

USB-C service and UART bridge



5.1 k Ω CC resistors

The connector exposes USB service power and data. USBLC6 protects the data pair.

CP2102N uses external 3.3 V

VDD and VREGIN share one rail in bypass mode. UART TX/RX use 100 Ω series resistors.

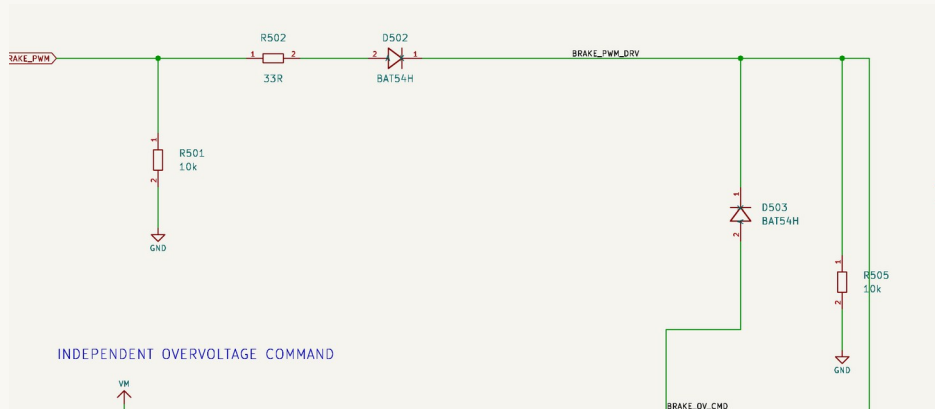
Source must permit ≥ 500 mA

There is no USB-PD negotiation or pre-enumeration/suspend load shedding. Reset and BOOT0 remain manual.

The 22.1 kΩ / 47.5 kΩ divider senses USB presence separately from bridge power.

Reset states and fault response

Event	Required response
Reset / erased MCU	Keep inverter PWM inactive
VM disappears / returns	Clear torque, reinitialize driver
Current / VDS fault	Latch off, investigate before rearm
Feedback / NTC invalid	Inhibit torque
Regeneration, supply cannot sink	Use qualified local energy dump



An unpopulated TVS or snubber footprint provides no fitted transient protection.

Inverter defaults inactive

Firmware must explicitly configure sensing, break paths and the driver before arming.

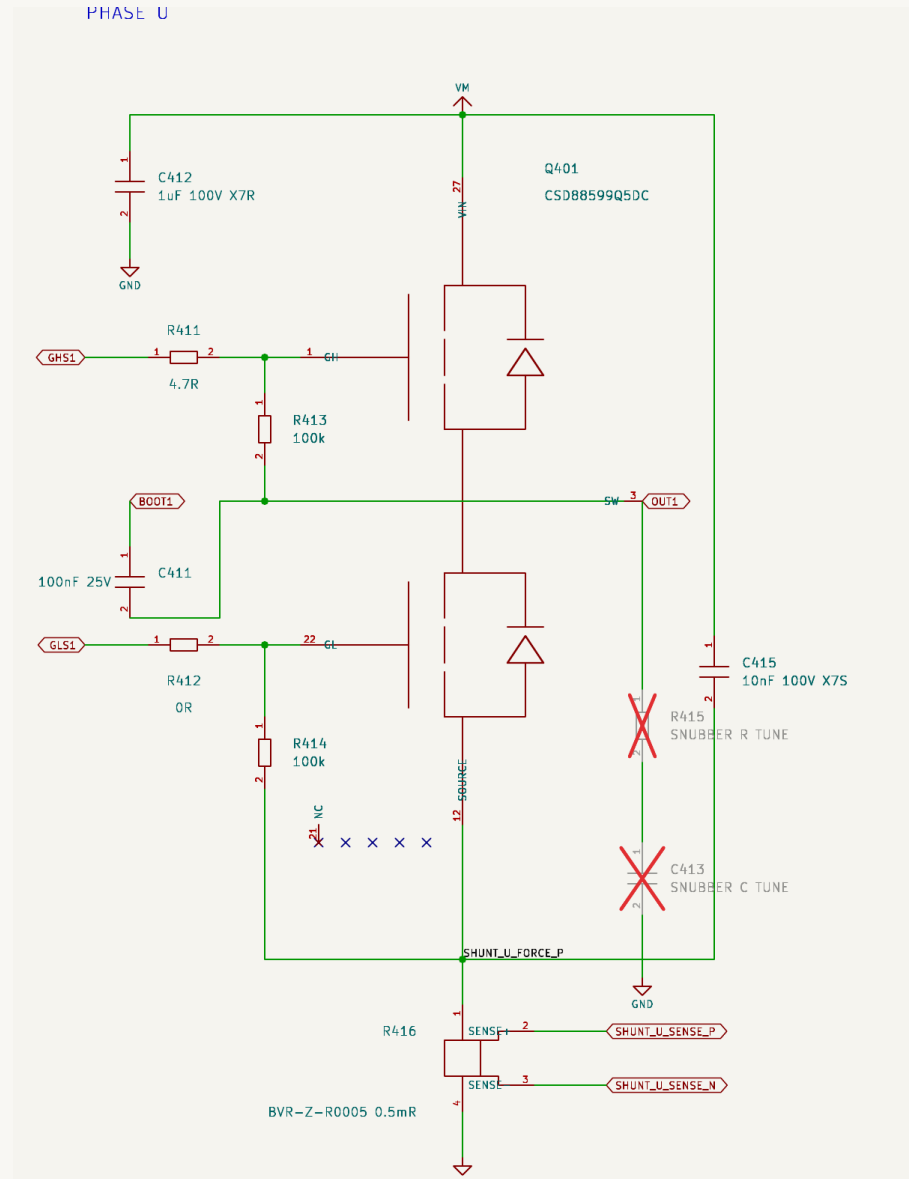
Different protections have dependencies

Timer-break comparison needs configuration. OV braking needs its supplies and a suitable external resistor.

A fault requires deliberate recovery

Latch inverter off, invalidate stale commands and recheck power, feedback and temperatures.

Layout requirements behind the schematic



Minimize the switching loop

Place local ceramics, MOSFETs and source return close together. Keep bootstrap and gate paths short.

Separate force and sense paths

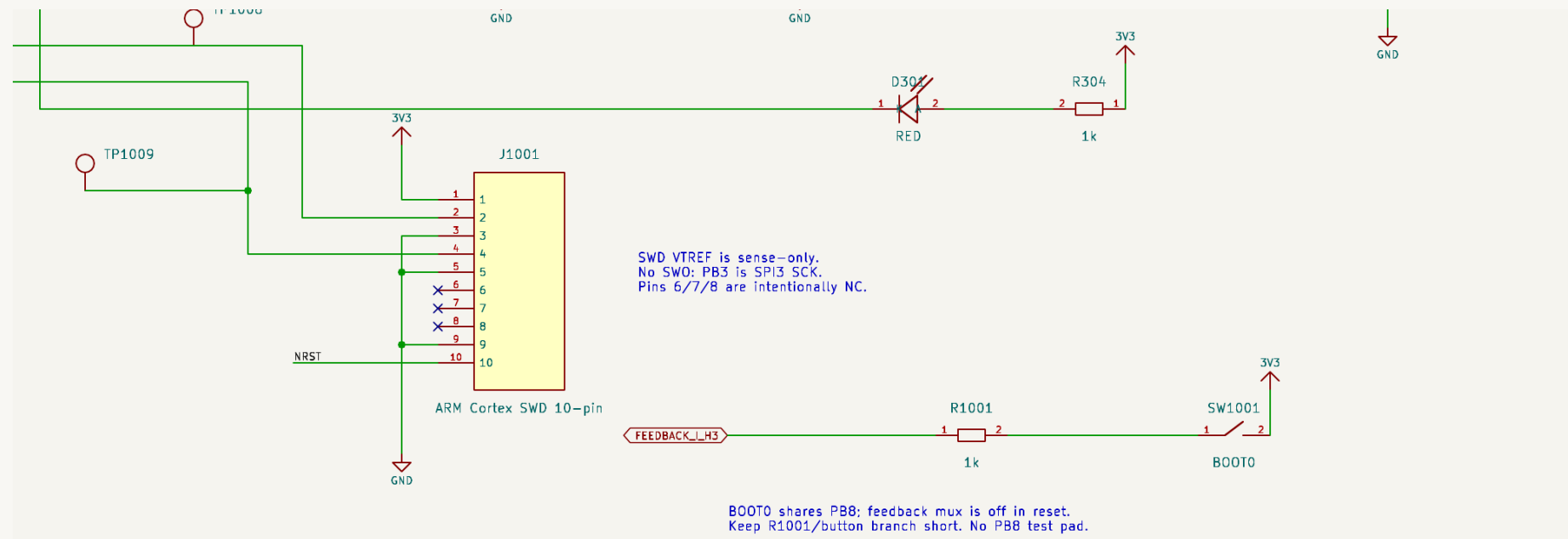
Kelvin pairs must not share motor-current copper. Keep analog nodes away from switch-node electric fields.

Thermal pads have electrical meaning

The CSD88 599 exposed VIN pad is live.
Q501 and LDO pads require their exact
package patterns.

Component values depend on the parasitic inductance and thermal paths created by the PCB.

Startup and low-energy commissioning



Verify logic power first

Check rails, source priority, backfeed and reset defaults. Program and retain recovery through SWD.

Prove protection before raising power

Calibrate ADCs and current offsets. Demonstrate timer break, VDS faults and hardware braking.

Begin unloaded and current-limited

18 V, 1–2 A current command, 20 kHz, 500 ns deadtime. Inspect gates and bootstrap voltage before increasing load.

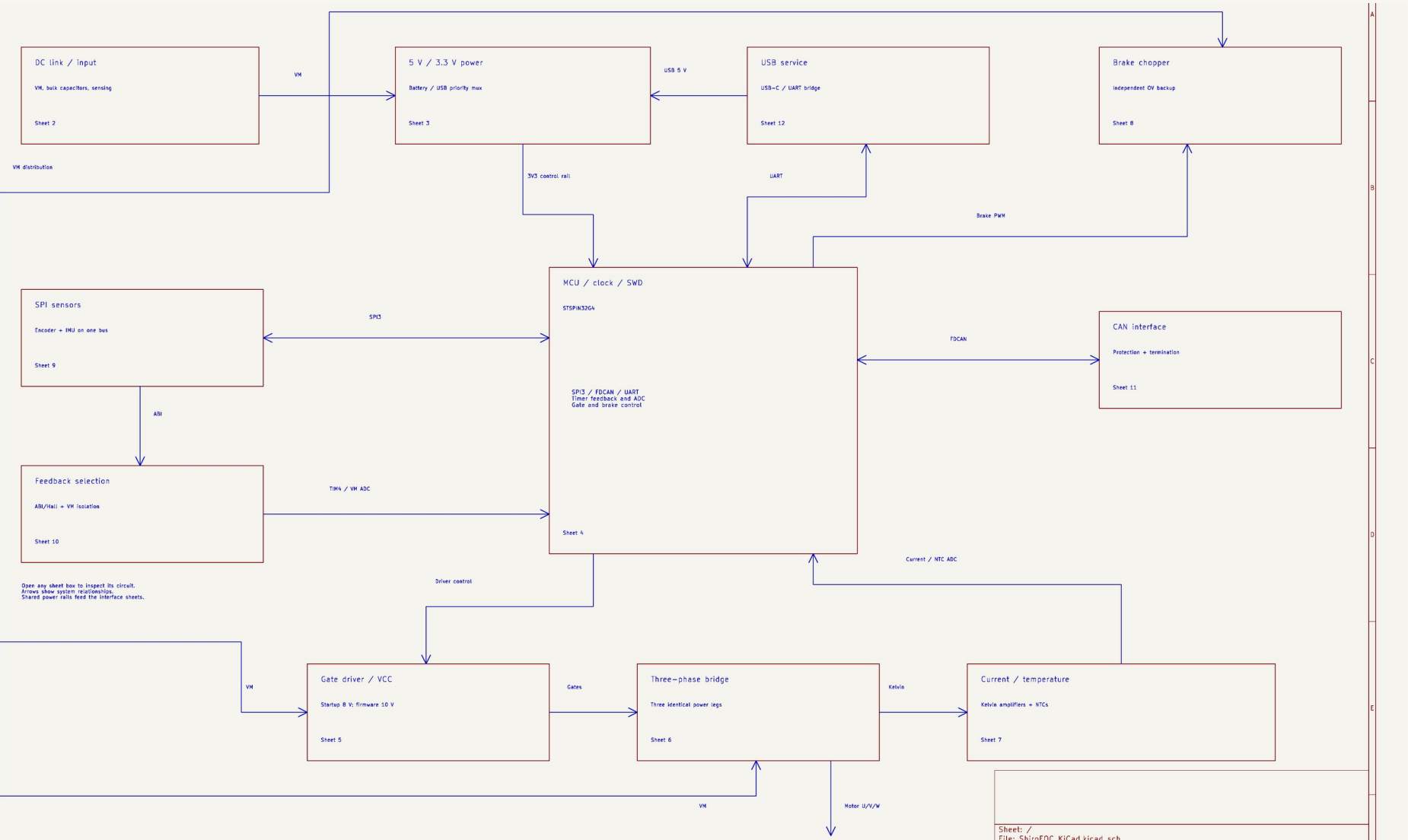
Voltage and current increase only after waveform, fault-response and thermal checks pass.

Choices requiring prototype measurements

Choice	Present value	Measurement that decides
Bootstrap / maximum duty	100 nF per phase	BOOT–OUT minimum and recharge
Gate resistance	4.7 Ω GH / 0 Ω GL	VGS, VDS, switching energy and heat
Phase snubber / input damping	DNP tuning positions	Ringling and hot-plug response
VCC inductor	22 μ H	Startup, regulation and ripple
Overcurrent threshold	Nominal +60 A	Calibrated trip and shutdown latency
DC-link bank	3 $\times$ 680 μ F	RMS ripple and capacitor temperature
Brake resistor	10 Ω example	Energy, repetition rate and cooling

Present values are starting selections. Measurements determine whether to keep or revise them.

Design conclusions and remaining decisions



The architecture is coherent

Integrated control, external power blocks, three shunts and local braking form a compact servo module.

The operating envelope is still conditional

Capacitor ripple, bridge overshoot and heat govern the useful continuous current and maximum bus.

Measurements close the design

Finalize gate timing, snubbers, brake energy and protection settings on the actual PCB.

Connectivity review supports layout. Hardware qualification establishes usable ratings.

A1. Bootstrap and gate-drive worksheet

Input / result	Value	Status or implication
Qg at 10 V	56 nC maximum	TI test conditions: 30 V, 30 A
Qgd / internal gate R	7 nC typ / 0.9 Ω typ	Use actual driver strength and waveform
CBOOT nominal / effective	100 nF / MPN dependent	Bias, temperature and tolerance remain inputs
Gate-only droop	0.56 V at 100 nF	1.12 V if effective capacitance is 50 nF
100 k Ω pull-down demand	100 μ A at 10 V	Adds 5 nC over 50 μ s
Charging resistance scale	100–200 Ω	RC = 10–20 μ s at 100 nF
90% empty-cap recharge	23–46 μ s, ideal RC	Startup illustration, not a PWM guarantee
Final acceptance inputs	UVLO, IHB, leakage, tON	Measure minimum BOOT–OUT and VGS

Charge balance and recharge time must both pass at the selected modulation limit.

A2. Power and analog calculation summary

Network	Calculation	Nominal result
5 V buck divider	$1 \times (1 + 100/24.9)$	5.016 V
Buck inductor ripple	$V_{OUT}(1 - V_{OUT}/V_{IN})/(L_f)$	0.41–0.50 App at 18–42 V
Mux OV threshold	$1.06 \times (21 + 5.10)/5.10$	5.43 V
LDO at 250 / 320 mA	$(5.25 - 3.3) \times I$	0.488 / 0.624 W
VM divider and filter	$V_M/21; (25.714k + 1k) \times 10n$	2.00 V at 42 V; 267 μs
Shunt at 40 / 80 A	$I^2 \times 0.5 \text{ m}\Omega$	0.8 / 3.2 W
Current gain / ideal code	$28 \times 0.5 \text{ m}\Omega; 3.3/4096/\text{gain}$	14 mV/A; 57.5 mA/code
SCREF and filter	$3.3/11; (100k 10k + 1k) \times 10n$	0.30 V; 101 μs

Nominal calculations use released values. Tolerance and operating conditions remain part of qualification.

A3. Brake threshold and energy worksheet

Quantity	Result	Interpretation
RU / RL / RF	450 kΩ / 12.7 kΩ / 1 MΩ	Reference and positive feedback set trips
Nominal rising / falling	45.92 V / 44.21 V	Includes project comparator model
Rising component corners	44.19–47.51 V	No dynamic bus overshoot included
Falling component corners	42.58–45.96 V	Do not assume fixed hysteresis at corners
10 Ω at 48 V	4.8 A; 230.4 W	Instantaneous fully-on stress
0.01 kg·m² at 2000 rpm	219.3 J	Rotational energy to zero speed
Example resistor requirement	≥250 W; 500 J over 5 s	Specified heatsink and pulse qualification
Missing resistor / missing rails	Braking unavailable	Installation and power dependencies

$$V_{M,trip} = V_{trip}(1 + R_U/R_L + R_U/R_F) - V_{OUT}(R_U/R_F)$$

A4. Schematic and component cross-reference

PDF page / module	Main slides	Key references
1 System overview	1, 3, 36	Root sheet
2 Battery / DC link	2, 5–7	J101, C101–C108, R101–R106
3 Auxiliary power	4, 8–10	U201, U203, U204, L201
4 Controller / debug	11, 34	U301 A/B, Y301, J1001
5 Driver / VCC	12, 17, 22	U301 C, L301, R310–R312
6 Inverter	13–19, 33, 35	Q401–Q403, C411/421/431
7 Current / temperature	20, 21, 23	U301 D/E/F, TH701–TH703
8 Brake	24–26, 32	U501, U502, Q501, J501
9 SPI sensors	27, 28	U601, U701
10 Feedback selection	7, 29	U602, J601, D601
11 CAN	30	U801, D801, J801/J802
12 USB service	31	U202, U901, J201

All 12 source sheets are represented. Full source citations and component detail are in the speaker notes.

A5. Sources and unresolved evidence

Project evidence

Current KiCad PDF/BOM, electrical calculations, design review, layout handoff and firmware contract.

Manufacturer evidence

STSPIN32G4, CSD88 599Q5DC and each supporting IC datasheet. Relevant sections and source paths accompany the slide notes.

Unresolved hardware evidence

Bootstrap margin, capacitor ripple, switch-node overshoot, protection calibration, temperatures and braking remain prototype tests.

No simulated or illustrative value in this deck represents a measured hardware result.